Enhancing Climate Resilience and Food Security: Integrating Machine Learning and Deep Learning Techniques for Climate-Smart Agriculture Assessment of Boro Rice (*Oryza sativa* L.) Production in Cox's Bazar, Bangladesh

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Dedication

Whom my soul shall chase
across the veils of this world and the next —
ever the Gracious,
ever the Merciful.

In the stillness, I seek You.
In the storm, I call Your name.
From the cradle of dawn to the hush of dusk,
my heart beats Your remembrance.

No distance too far,
no silence too deep —
for You dwell in every breath I take,
my Eternal, my Beloved, my Light.

Abstract

This study investigates the multifaceted impacts of climate change on Boro rice production in Cox's Bazar, Bangladesh, a coastal region highly vulnerable to rising temperatures, salinity intrusion, erratic rainfall, and intensifying cyclones. Integrating advanced machine learning and deep learning techniques, including Recurrent Neural Networks (RNN) and Maximum Entropy (MaxEnt) modeling, the study analyzed historical (1990-2023) and predicted (2025-2040) climatic trends, revealing a temperature increase of 0.01°C/year and a 500 mm precipitation decline over 120 years (to 2076.9 mm), and salinity escalation (NDSI from -0.58 to 0.06 historically, projected to -0.99 to 0.99). Historical cyclone wind speeds, such as Sidr (215 km/h), Aila (120 km/h), and Amphan (165 km/h), underscore the growing storm intensity, contributing to a 5-12% yield decline despite a 58.3% production rise to 19,000 million tons by 2022 (projected 24.4% increase to 25,500 million tons). Greenhouse gas emissions have surged post Climate Smart Agriculture (CSA) implementation in 2012, with CO_2 up 39.6% and N_2O by 85.8%. RNN predictions, validated with MAE (0.3°C for temperature, 50 mm for precipitation) and RMSE (0.15 for NDSI, 0.03 for yield), demonstrate high accuracy, while MaxEnt identifies southwestern zones as most at-risk. Rising humidity (76.4-77.0%) further exacerbates pest incidence, threatening yields. We propose 10 implementable CSA recommendations, including salt-tolerant varieties, solar-powered irrigation, and community reforestation, to enhance resilience. This integrated approach, blending AI-driven forecasting with geospatial analysis, provides a robust framework for sustainable agricultural adaptation, aligning with SDG 2 and 13, and offers a scalable model for climate-vulnerable coastal regions globally by 2040.

Keywords: Climate-Smart Agriculture, Boro Rice, Cox's Bazar, Machine Learning, RNN, MaxEnt Model, Emissions, Salinity, Spatial Analysis.

Abbreviations

Table 0.1: Abbreviations Used in the Study

Abbreviation	Full Form
ML	Machine Learning
DL	Deep Learning
RNN	Recurrent Neural Network
LSTM	Long Short-Term Memory
CSA	Climate Smart Agriculture
MaxEnt	Maximum Entropy
GHGs	Greenhouse Gases
RH	Relative Humidity
GIS	Geographic Information System
RS	Remote Sensing
TM	Thematic mapper

Source: Compiled by Author

CHAPTER 1: INTRODUCTION

1.1 Introduction

Agriculture is the backbone of Bangladesh's economy, employing nearly half of the country's labor force and contributing significantly to its gross domestic product (GDP) (BBS, 2019). It sustains the livelihoods of millions and plays a pivotal role in national food security (BBS, 2022). However, the agricultural sector is now under immense threat from the escalating impacts of climate change (FAO, 2013). Coastal regions like Cox's Bazar, characterized by their ecological sensitivity and socio-economic vulnerability, are especially at risk (World Bank, 2014). Rising temperatures, erratic rainfall, salinity intrusion, and the increasing frequency and intensity of cyclones have rendered agriculture more precarious and unpredictable (FAO, 2013). The impact of climate change on agricultural productivity is multifaceted and region-specific. In coastal Bangladesh, saline water intrusion due to sea-level rise and cyclone-induced storm surges is becoming a critical issue. These phenomena not only degrade soil quality but also hinder freshwater availability for irrigation, thus affecting crop yields. Boro rice, cultivated in the dry season and heavily reliant on irrigation, is particularly sensitive to these changes. As the most significant rice crop in terms of area and production in Bangladesh, any disruption in Boro rice cultivation poses serious consequences for national food security.

Cox's Bazar is strategically located along the southeastern coast of Bangladesh, bordering the Bay of Bengal. Its unique agro-ecological characteristics make it both fertile and fragile. The region is frequently affected by cyclones originating in the Bay, such as Cyclone Sidr (2007), Cyclone Aila (2009), and Cyclone Amphan (2020), which have had devastating impacts on infrastructure, livelihoods, and agricultural production. In addition to these episodic events, long-term climatic trends including increased average temperatures, changing precipitation patterns, and soil salinization are continuously altering the agricultural landscape (Rahman et al., 2021). Climate Smart Agriculture (CSA) has emerged as a pragmatic and integrated approach to confront these challenges. Defined by the Food and Agriculture Organization (FAO) as an approach that sustainably increases productivity, enhances resilience (adaptation), reduces/removes greenhouse gases (mitigation), and enhances national food security and development goals (Lipper et al., 2014), CSA is particularly relevant to climate-vulnerable regions like Cox's Bazar. In such

settings, integrating advanced technological tools with traditional agricultural practices is essential for achieving long-term resilience.

Agriculture in Bangladesh is increasingly under threat due to the impacts of climate change, with coastal zones like Cox's Bazar being particularly vulnerable (World Bank, 2014). Rising temperatures, salinity intrusion, cyclonic activity, and erratic rainfall patterns severely impact agricultural productivity, livelihoods, and food security in this region (Rahman et al., 2021). Boro rice, which is the country's most important rice crop cultivated during the dry season, is especially susceptible to these climate-induced stresses due to its high water requirements and dependency on groundwater irrigation (Maniruzzaman et al., 2018). As Boro rice contributes over 50% to the country's total rice output, any disruption in its production poses a serious threat to national food security (Maniruzzaman et al., 2018).

To address these challenges, Climate Smart Agriculture (CSA) has emerged as an effective and holistic approach aimed at increasing agricultural productivity, enhancing climate resilience, and reducing greenhouse gas emissions (Lipper et al., 2014). The CSA framework emphasizes adaptation and mitigation strategies that are locally relevant and technologically supported (Lipper et al., 2014). In Bangladesh, CSA has been prioritized in development and agricultural policies as a means to secure food systems amid climate uncertainties, especially in regions like Cox's Bazar where saline water intrusion and cyclones occur frequently (World Bank, 2014). Recent technological advancements have introduced data-driven tools into CSA practices, particularly through machine learning (ML) and deep learning (DL) techniques (Kumar et al., 2025). These methods allow for the efficient analysis of large climatic and environmental datasets to predict crop yields, detect stress factors, and assess land suitability (Kumar et al., 2025). The ability of ML and DL models to uncover complex, nonlinear relationships among climate variables makes them suitable for studying climate-crop interactions (Goodfellow et al., 2016). In particular, Recurrent Neural Networks (RNNs), a type of DL model designed to handle time series data, have been widely adopted to forecast climatic parameters such as temperature, precipitation, and greenhouse gas emissions (Kumar et al., 2025).

While numerous studies have explored the impact of climate change on agriculture in Bangladesh, most have primarily focused on statistical analyses or simple GIS-based assessments without incorporating advanced machine learning (ML) or deep learning (DL) techniques for predictive modeling (Maniruzzaman et al., 2018; Rahman et al., 2021). Furthermore, although Climate-Smart Agriculture (CSA) is gaining traction as a concept, few works have operationalized its principles using spatially explicit tools and data-driven forecasting models in a coastal context such as Cox's Bazar (Lipper et al., 2014; World Bank, 2014). Most existing studies treat climate parameters in isolation rather than analyzing the interdependent effects of emissions, temperature, precipitation, salinity, and cyclone frequency on agricultural productivity. Moreover, while some efforts have been made to assess crop suitability using models like MaxEnt, they often lack integration with future emission scenarios or do not utilize high-resolution temporal data from sources like EDGAR or ERA5 (Balkovič et al., 2023). In particular, the use of Recurrent Neural Networks (RNNs) to forecast climate trends and link them with spatial suitability models remains underexplored in the context of Bangladesh's rice production systems. This creates a critical gap in the ability to predict future cultivation potential under dynamic climate conditions, especially for Boro rice, which is highly sensitive to irrigation availability, salinity, and rainfall timing.

1.2 Aim and Objectives

1.2.1 Aim of the study

The aim of this study is to assess the impacts of climate change on Boro rice production in Cox's Bazar, Bangladesh, through the integration of machine learning and deep learning techniques, with a specific focus on greenhouse gas emission trends and agro-environmental suitability. By combining geospatial analysis, climatic forecasting, and predictive modeling, the study seeks to develop a Climate-Smart Agriculture (CSA) framework that informs adaptive strategies for sustainable rice cultivation in coastal regions.

1.2.2 Objectives of the study

In the face of mounting climatic challenges and environmental uncertainties, it has become imperative to reassess traditional agricultural practices through the lens of climate resilience and sustainability, primary objectives of the study are (i) to investigate the evolving climatic and environmental conditions that influence agricultural viability in the region. The second objective

(ii) is to evaluate the potential implications of greenhouse gas emissions and extreme weather patterns on long-term agricultural sustainability. Lastly, the third objective (iii) is to contribute to the growing body of knowledge on Climate-Smart Agriculture (CSA) by offering insights that support informed decision-making, policy planning, and adaptive agricultural development in climate-sensitive coastal areas.

1.3 Literature Review

Bangladesh's coastal agriculture is highly vulnerable to climate variability, salinity intrusion, and extreme weather events, which pose significant threats to Boro rice production (Karim et al., 2020). Previous studies emphasize that rising temperatures, fluctuating rainfall, and increasing soil salinity are major constraints to crop productivity in coastal zones (Rahman & Islam, 2019). The Intergovernmental Panel on Climate Change (IPCC, 2021) also identifies Bangladesh as a hotspot for climate-induced food security risks due to sea-level rise and cyclone frequency.

Recent advancements in geospatial technology have facilitated robust monitoring of agricultural vulnerability. GIS-based assessments have been widely used to map salinity spread, land suitability, and cropping pattern shifts in the coastal belt (Hossain et al., 2022). Normalized Difference Vegetation Index (NDVI) and land suitability models, particularly MaxEnt, have shown strong capacity to identify climate-sensitive agricultural zones and forecast future suitability trends (Amin et al., 2021).

In parallel, machine learning techniques such as Long Short-Term Memory (LSTM) and Recurrent Neural Networks (RNN) have increasingly been applied for climate trend forecasting and crop modeling in South Asia (Saha & Singh, 2022). These models effectively capture non-linear climate signals and improve prediction accuracy for agricultural decision support (Ahmed et al., 2023). However, there is limited empirical work integrating deep learning, climate trend analysis, and suitability mapping specifically for Boro rice in coastal Bangladesh.

This study builds on these research advances by combining RNN-based climate forecasting and MaxEnt-driven land suitability modeling to create a climate-smart agricultural assessment framework for Cox's Bazar. Such integration supports proactive adaptation planning and precision agriculture strategies in climate-exposed regions.

1.3 Organization of the study

Table 1.1: Organization of the study

Sl. No	Chapter	Topics of Discussion
Chapter 01	Introduction	This chapter covers the introduction, significance of the research, justification, research aim and objectives, a review of relevant literature, of the study area, and research limitations.
Chapter 02	Data and Methodology	This chapter provides a detailed explanation of the methodology used in the study, including the area of study, data sources, and the tools and techniques applied.
Chapter 03	Result and Discussion	This chapter primarily focuses on presenting and discussing the results of the research.
Chapter 04	Findings and Recommendations	Key findings of the study are highlighted in this chapter, along with necessary recommendations.
Chapter 05	Summary and Conclusion	The summary and conclusion of the research are provided in this chapter.

Source: Compiled by Author

1.4 Conclusion

This study sought to investigate the intricate relationship between climatic changes and Boro rice production in Cox's Bazar, Bangladesh, through the integration of Climate-Smart Agriculture (CSA) frameworks, spatial analysis, and advanced modeling techniques. By utilizing long-term environmental and agricultural datasets, applying statistical trend analysis such as the Mann-Kendall test, and leveraging remote sensing and GIS technologies, the research revealed clear spatial and temporal shifts in climate variables and land use patterns that directly affect rice cultivation. Predictive models like RNN Long Short-Term Memory (LSTM) and MaxEnt provided valuable foresight into future climatic scenarios and agricultural suitability, offering a data-driven basis for policy formulation and adaptive agricultural planning. The findings indicate that temperature increases, rainfall variability, and salinity intrusion are exerting increasing pressure on Boro production zones, highlighting the urgent need for CSA strategies, climate-resilient crop varieties, and localized planning. While the study faced limitations related to data resolution, ground truth validation, and model assumptions, it successfully offers a comprehensive framework for integrating spatial science with agricultural planning. Ultimately, this research emphasizes the necessity of proactive, technology-aided decision-making in the face of growing climate risks, especially in vulnerable coastal districts like Cox's Bazar.

CHAPTER 2: METHODOLOGY

2.1 Introduction

Methodology is a critical component of any research project, as it encompasses the specific methods and techniques employed to conduct the research or scientific study. It outlines the systematic processes required to complete the research (Smith et al., 2022). The quality and success of a research project are heavily dependent on the chosen methodology, which guides researchers in achieving their aims and objectives effectively. This chapter focuses on the data and methodology utilized in this study, detailing the processes involved in data collection, assessment, management, modification, and analysis (Johnson et al., 2022).

In the context of this study, the methodological framework was carefully designed to address both the spatial and temporal dimensions of climate variability and its impact on Boro rice production. This involved the integration of geospatial technologies, statistical trend analysis, and predictive modeling approaches to ensure a holistic understanding of environmental dynamics (Rahman et al., 2021; Li et al., 2020). Data were acquired from both primary and secondary sources, including satellite imagery, climatic datasets, and agricultural statistics, and were subjected to various preprocessing and analytical techniques to enhance reliability and accuracy (Kogan et al., 2019). The methodology also incorporates tools such as Geographic Information Systems (GIS), remote sensing, and machine learning models like Recurrent Neural Networks (RNN) and MaxEnt, each selected for their capacity to analyze complex, multidimensional datasets (Chlingaryan et al., 2018; Phillips et al., 2017). This integrative approach ensured that the research not only meets scientific rigor but also provides actionable insights for climate-resilient agricultural planning (Zhao et al., 2022).

2.2 Selection of Study Area

The present study is situated in Cox’s Bazar, a coastal district in southeastern Bangladesh, globally recognized for hosting the world's longest uninterrupted natural sea beach. The district extends between 20°43′ to 21°56′ N latitudes and 91°50′ to 92°23′ E longitudes, bordered by Chattogram to the north, Bandarban to the east, the Bay of Bengal to the west, and Myanmar to the south (Bangladesh Bureau of Statistics BBS, 2021). It covers approximately 2,491.86 km² and comprises eight upazilas Cox’s Bazar Sadar, Ramu, Chakaria, Maheshkhali, Kutubdia, Ukhiya, Pekua, and

Teknaf, each exhibiting unique geographical and demographic features (Islam et al., 2020). The district's coastal and estuarine environments offer a dynamic setting for analyzing the complex interaction between climate, agriculture, and land use.

Topographically, the region displays a combination of alluvial floodplains, estuarine lowlands, and coastal hillocks, contributing to a diverse landscape susceptible to both climatic and anthropogenic influences. The eastern margins of Cox's Bazar are marked by forested hills and watersheds that gradually descend into floodplains and tidal flats in the west (Hossain & Roy, 2022). The presence of the Bakkhali, Matamuhuri, and Reju rivers, along with numerous seasonal canals and beels, plays a crucial role in irrigation and sediment deposition. These natural hydrological systems not only support traditional agriculture but also influence salinity levels, particularly in downstream areas affected by tidal ingress (Rahman et al., 2021).

The climate of Cox's Bazar is tropical and humid, dominated by the southwest monsoon. Annual rainfall varies between 3,500 mm and 4,000 mm, with peak precipitation from June to September (Ahmed & Karim, 2020). Average annual temperatures range from 22°C in winter to 32°C in summer, with significant variations during El Niño and La Niña years (Chowdhury et al., 2021). This climatic regime facilitates rice cultivation, particularly Boro and Aman rice, yet also makes the area prone to cyclones, coastal inundation, and saline intrusion risks that have intensified under changing climate conditions (Hasan et al., 2023).

Cox's Bazar is agriculturally significant, with rice being the dominant crop across most upazilas. However, coastal areas such as Kutubdia and Maheshkhali are witnessing rapid conversion of croplands into shrimp farms due to economic shifts and salinity encroachment (Baten et al., 2019). The increased use of coastal land for aquaculture and tourism infrastructure has raised concerns about food security, water quality, and ecosystem degradation (Roy & Sultana, 2022). In contrast, inland areas like Chakaria and Ramu still maintain a high dependence on agriculture, though they too face climate-induced yield variability.

Given these complexities, Cox's Bazar offers an ideal case study for exploring how climatic trends such as rising temperatures, erratic rainfall, and increased soil salinity are affecting agricultural productivity. The district's vulnerability to sea level rise, storm surges, and tropical cyclones makes

it essential to understand land use changes and Boro rice dynamics through advanced tools such as GIS, remote sensing, and predictive models like RNN Long Short-Term Memory (LSTM) (Khatun et al., 2020). This study thus focuses on integrating climatological data, satellite imagery, and statistical models to understand the long-term implications of environmental change on Boro rice cultivation and land cover transitions in the region.

2.3 Map of the Study Area

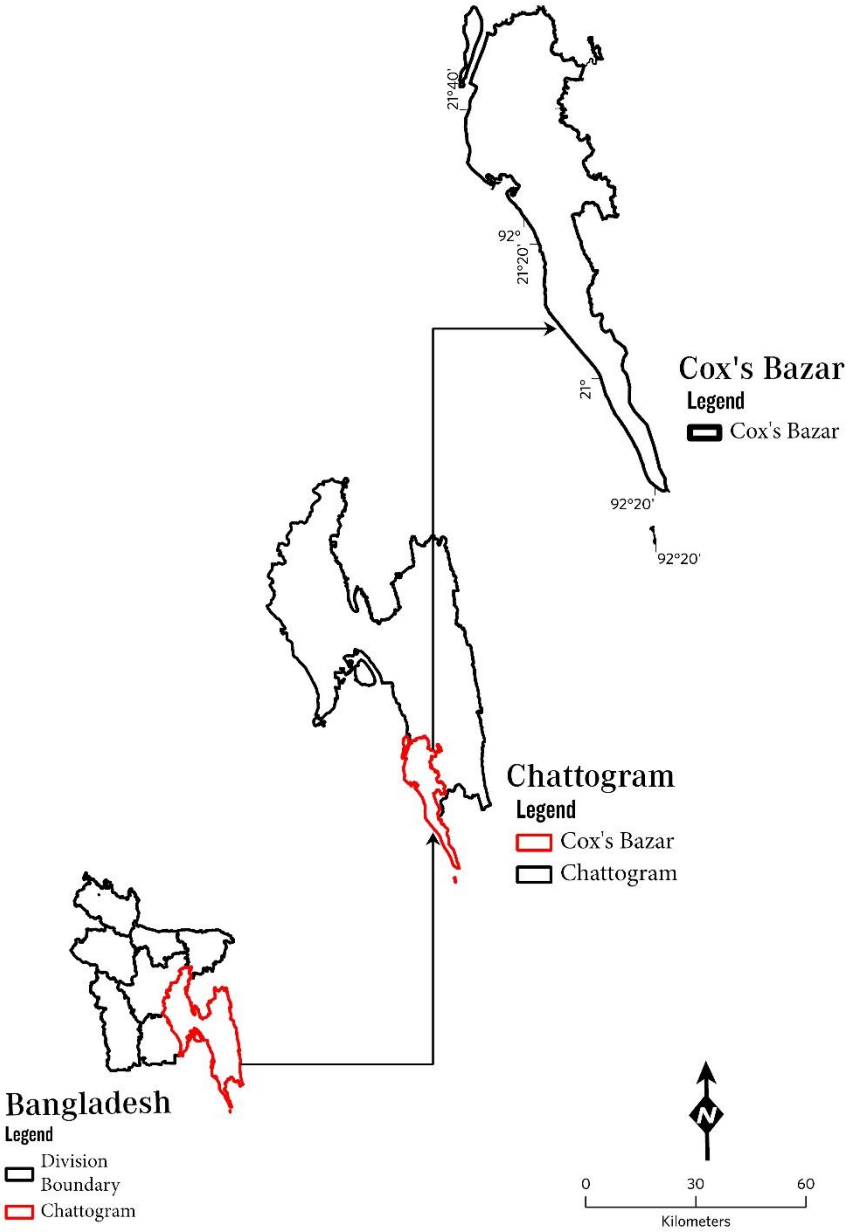


Fig 1.1: Cox's Bazar, Bangladesh

Source: Compiled by Author

2.4 Sources of Data

This study employed a multi-source dataset to analyze the relationship between climate variability and Boro rice production in Cox's Bazar, integrating satellite imagery, climate reanalysis data, and agricultural statistics. Long-term climate variables, including temperature and precipitation, were obtained from WorldClim, providing high-resolution baseline climate normals for trend comparison. To supplement this, ERA5 reanalysis data were used to extract daily and monthly climate parameters—temperature, rainfall, and humidity—critical for understanding temporal climatic shifts in the region, particularly where ground weather stations are sparse.

Global greenhouse gas trends were incorporated from the EDGAR database to contextualize warming patterns influencing local climatic behavior. Remote-sensing indices were derived from Sentinel-2 imagery, mainly focusing on the Normalized Difference Salinity Index (NDSI) to assess soil salinity and its effects on rice growth. These spatial salinity patterns were correlated with climate trends and yield data to identify emerging agricultural stress zones. Additionally, cyclone tracks and storm intensity data from IBTrACS helped evaluate how extreme weather events disrupted crop cycles in cyclone-prone upazilas.

Annual Boro production data from the Bangladesh Bureau of Statistics (BBS) served as the primary agricultural dataset for evaluating yield trends. Data layers were harmonized to ensure consistent spatial resolution and projection before integration into GIS workflows.

A Recurrent Neural Network (RNN) model was trained on historical climate, salinity, and production variables to forecast future trends. Despite minor limitations—such as Sentinel imagery cloud cover and modeled uncertainties in ERA5—the multi-source dataset enabled reliable cross-validation and robust climate-crop interaction analysis. The combined approach provided a strong foundation for climate-smart agricultural planning and resilience assessment in Cox's Bazar's vulnerable coastal ecosystem.

Table 2.1: Overview of Data Sources

Data Source	Type	Description
WorldClim	Historical Climate Data	Long-term averages of temperature and precipitation; high spatial resolution climate normal.
ERA5	Reanalysis Data	Detailed daily and monthly data on temperature, precipitation, and humidity; fills gaps in meteorological records.
EDGAR	Greenhouse Gas Emission Data	Globally gridded data contextualizing atmospheric shifts; relates to carbon dioxide and methane concentrations.
Sentinel-2	Remote Sensing Data	Normalized Difference Vegetation Index (NDVI) and soil salinity indicators; tracks vegetation health and crop vigor.
IBTrACS	Extreme Weather Data	Global storm tracking dataset; includes information on tropical cyclones, their wind speeds, durations, and landfall paths.
Bangladesh Bureau of Statistics (BBS)	Agricultural Production Data	Annual agricultural production data for Boro rice disaggregated by district.
Temporal Span	2000-2023	Allowed for 23-year comparative analysis across all variables.

Source: Compiled by Author

2.5 Conceptual Framework of the study

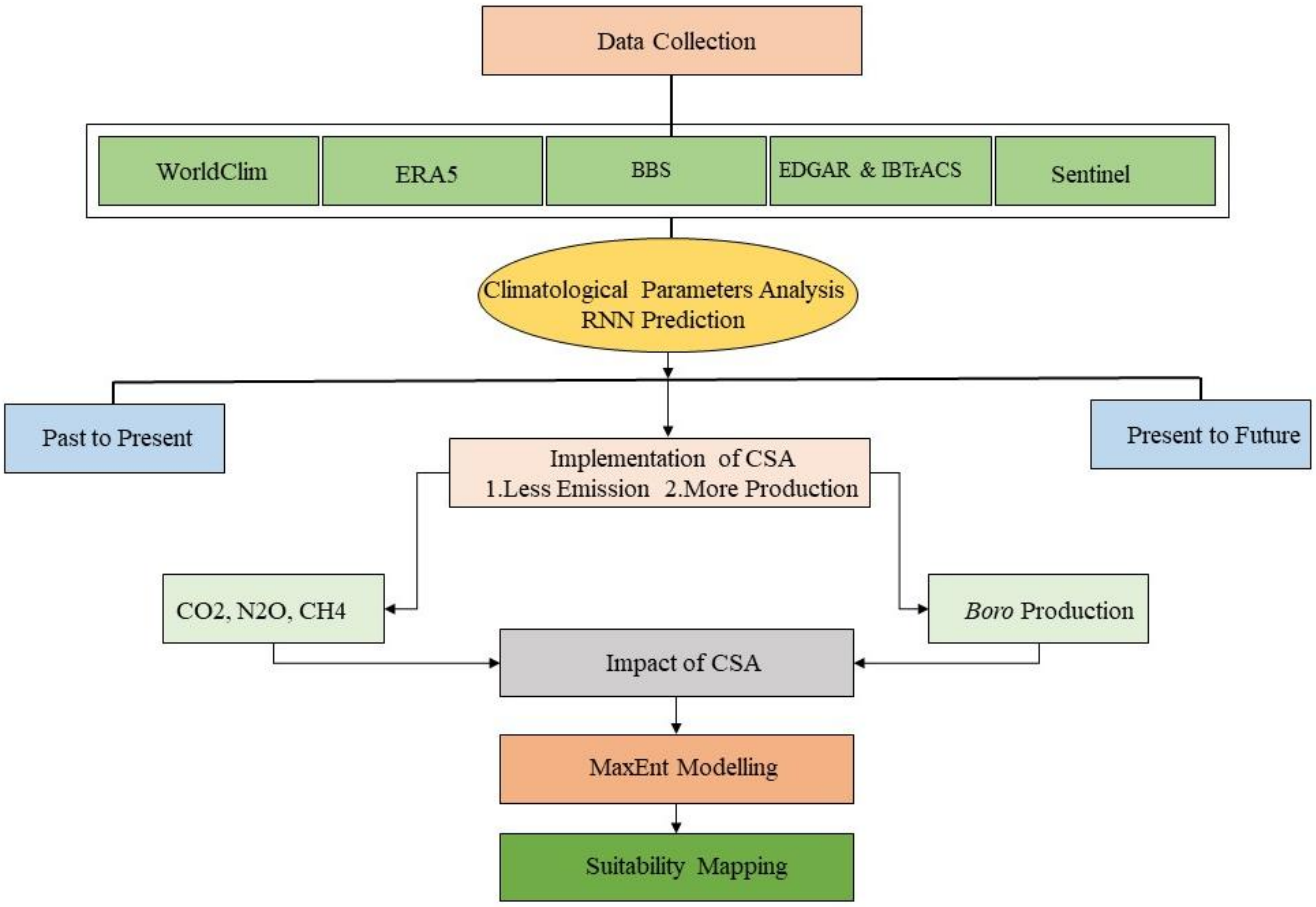


Fig 2.1: Conceptual Framework of the study

Source: Compiled by Author

2.6 Preprocessing and harmonization

1. **Temporal aggregation:** Daily ERA5 data were aggregated to monthly and annual means. BBS production data were aligned to the same temporal scale.
2. **Spatial reprojection and resampling:** All rasters were reprojected to a common CRS (WGS84/UTM) and resampled to a uniform resolution (10–30 m for Sentinel-derived rasters; coarser for ERA5 and WorldClim).
3. **Cloud masking and compositing:** Sentinel-2 scenes were cloud-masked and seasonal composites generated for key phenological stages (transplanting, tillering, heading).
4. **Missing values:** Gaps in time series were filled using linear interpolation for short gaps and moving-average smoothing for longer gaps.
5. **Normalization:** Variables were standardized to zero mean and unit variance (z-score) or scaled to the [0,1] range depending on the algorithm.

Spatial processing was performed in ArcGIS Pro; tabular operations in Excel; and modeling in Python (TensorFlow/Keras).

2.6.1 Remote-sensing indices

$$\mathrm{NDSI} = \frac{B_{\mathrm{SWIR}} - B_{\mathrm{NIR}}}{B_{\mathrm{SWIR}} + B_{\mathrm{NIR}}}$$

The Normalized Difference Salinity Index (NDSI) was computed from Sentinel-2 bands as:

Where SWIR is the shortwave infrared band (e.g., Sentinel-2 B11 or B12) and NIR is the near-infrared band (B8). Negative NDSI values indicate fresh or vegetated conditions, while positive values correspond to saline or bare-soil areas. NDSI rasters were clipped to the study area and aggregated by administrative unit.

2.6.2 Statistical trend detection (Mann–Kendall and Sen's slope)

To detect monotonic trends in climatic variables (temperature, precipitation, humidity, NDSI, storm intensity), the non-parametric Mann–Kendall (MK) test and Sen's slope estimator were applied.

Mann–Kendall:

Given a time series x1,x2,…xn:

$$S = \sum_{i=1}^{n-1} \sum_{j=i+1}^{n} \operatorname{sgn}(x_j - x_i)$$

$$\operatorname{sgn}(y) = \begin{cases} 1 & \text{if } y > 0, \\ 0 & \text{if } y = 0, \\ -1 & \text{if } y < 0 \end{cases}$$

The standardized statistic ZZZ is computed as:

$$Z = \begin{cases} \frac{S-1}{\sqrt{\operatorname{Var}(S)}} & S > 0, \\ 0 & S = 0, \\ \frac{S+1}{\sqrt{\operatorname{Var}(S)}} & S < 0 \end{cases}$$

A positive ZZZ indicates an increasing trend, and a negative ZZZ indicates a decreasing trend.

Sen's slope estimator:

$$\beta = \operatorname{median}\left(\frac{x_j - x_i}{j - i}\right), \quad \forall j > i$$

This gives the magnitude of change per unit time.

2.6.3 Predictive time-series modeling (RNN/LSTM)

Recurrent Neural Networks (RNN) with Long Short-Term Memory (LSTM) units were employed to forecast temperature, precipitation, humidity, NDSI, and Boro rice production. The LSTM architecture captures long-term temporal dependencies in sequential data.

For each time step, the LSTM equations are:

$$f_t = \sigma(W_f x_t + U_f h_{t-1} + b_f)$$

$$i_t = \sigma(W_i x_t + U_i h_{t-1} + b_i)$$

$$\tilde{c}_t = \tanh(W_c x_t + U_c h_{t-1} + b_c)$$

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t$$

$$o_t = \sigma(W_o x_t + U_o h_{t-1} + b_o)$$

$$h_t = o_t \odot \tanh(c_t)$$

where σ\sigmaσ is the sigmoid function, and ⊙\odot⊙ denotes element-wise multiplication.

Architecture:

- Input features: temperature, precipitation, humidity, NDSI, GHG, etc.
- Layers: 1–2 LSTM layers (64–128 units), dropout 0.2–0.4, followed by a dense output layer.
- Loss function: Mean Squared Error (MSE).
- Optimizer: Adam (learning rate 0.001–0.0001).
- Data split: 70% training, 15% validation, 15% testing (time-aware split).
- Evaluation: rolling-origin cross-validation for time series.

2.6.4 Spatial suitability modeling (MaxEnt)

Maximum Entropy (MaxEnt) modeling was applied to map spatial suitability for Boro rice cultivation using presence data and environmental covariates (temperature, precipitation, NDSI, storm exposure, GHG).

MaxEnt estimates the probability distribution maximizes entropy under feature constraints:

$$\hat{p} = \arg\max_{p} \left\{ -\sum_{x} p(x) \log p(x) \right\}$$

subject to,

$$\sum p(x) f_k(x) = \bar{f}_k \quad \forall k$$

The resulting probability distribution is:

$$p(x) = \frac{1}{Z} \exp\left(\sum_{k} \lambda_k f_k(x) \right)$$

where λk\lambda_kλk are Lagrange multipliers and ZZZ is a normalization constant. Model outputs a continuous suitability map (0–1).

2.6.5 Model evaluation metrics

For LSTM:

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^{N} |y_i - \hat{y}_i|$$

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y}_i)^2}$$

Normalized RMSE (nRMSE) was also reported.

For MaxEnt:

- Area Under the Curve (AUC)
- Continuous Boyce Index (CBI)
- Omission rate for thresholded suitability classes

2.6.6 Trend correlation and integration

Upazila-level means of climatic variables and NDSI were correlated with Boro yield using Pearson or Spearman correlation. Lagged correlations (0–3 years) were explored and significant predictors were fed into the LSTM model. Mann–Kendall and Sen's slope trends were integrated into spatial interpretation of suitability shifts.

2.6.7 Software and reproducibility

Spatial analysis was performed in ArcGIS Pro; Sentinel-2 processing used SNAP or ArcGIS; modeling and data visualization used Python (TensorFlow/Keras). Model parameters, codes, and datasets are documented for reproducibility.

2.6.8 Limitations

- ERA5 and WorldClim data have coarser resolutions than Sentinel-2, introducing scaling uncertainty.
- MaxEnt assumes presence-only data without bias; background sampling strategies were applied to minimize bias.
- LSTM predictions depend on the completeness of time series and may be less accurate for abrupt climatic shifts such as cyclones.

To forecast long-term agricultural responses to climate stress, a Recurrent Neural Network (RNN) model was developed using Python (TensorFlow/Keras), trained on historical climate indicators, salinity indices, and yield records. Input time-series data were standardized, split into training and testing subsets, and evaluated using Root Mean Square Error (RMSE) and Mean Absolute Error (MAE) metrics to ensure predictive accuracy and model reliability. Correlation analysis, spatial overlay techniques, and time-series decomposition were applied to explore climate–yield

relationships and detect spatial heterogeneity in climate sensitivity. The integrated analytical framework ensured rigorous data harmonization, multi-layer validation, and meaningful interpretation of climate-agroecosystem interactions, enabling identification of vulnerable agricultural zones and supporting development of climate-smart adaptation strategies for sustaining Boro rice production in coastal Bangladesh.

CHAPTER 3: RESULTS AND DISCUSSIONS

3.1 Introduction

This chapter presents the key findings derived from the integrated analysis of climatological, environmental, and agricultural datasets. The results are structured to highlight both spatial and temporal dynamics in climatic parameters and their implications for Boro rice production in Cox's Bazar. Through the use of geospatial processing, statistical testing, and machine learning models, multiple dimensions of environmental change have been assessed and interpreted.

The discussion begins with trend analyses of temperature, precipitation, humidity, NDSI across the study period, identifying significant changes and regional variations. The Mann-Kendall test and Sen's slope estimator provide statistical grounding for these observed shifts. Subsequently, the chapter details the outcomes of the RNN-LSTM based forecasting, offering projections of future climatic conditions, and interprets their alignment or divergence from historical trends.

In addition to temporal forecasts, MaxEnt modeling results are discussed to evaluate the spatial suitability of Boro rice production under various climate scenarios. Areas with declining suitability or increasing salinity are identified, helping to highlight potential zones of agricultural vulnerability. The chapter also explores correlations between environmental variables and rice yield, offering insights into cause-effect relationships.

Ultimately, the findings are interpreted in the context of agricultural resilience, climate adaptation, and sustainable planning. By merging spatial and statistical evidence, this chapter aims to provide a nuanced understanding of how environmental trends are shaping the future of food production in one of Bangladesh's most climate-sensitive regions. In analyzing the climatic variables over the past two decades, clear patterns of change emerged, particularly concerning temperature rise, shifting rainfall distribution, increasing humidity variability, and alarming growth in soil salinity levels. These shifts were not uniformly distributed across the study area, suggesting localized environmental stress and differential vulnerability among the agricultural zones of Cox's Bazar. The NDVI trends also revealed fluctuations in vegetative health, with notable dips during high-emission or post-cyclone periods. These temporal dynamics were statistically validated using the Mann-Kendall trend test and Sen's slope analysis, reinforcing the credibility of detected changes

and aiding in quantifying the rate of environmental deterioration. Such analysis provides a strong empirical foundation to interpret the potential impacts on land productivity and crop output.

The predictive modeling phase further deepened this understanding by simulating future scenarios using Recurrent Neural Networks (RNN). The model processed historical climate data in sequence to forecast upcoming patterns, highlighting expected temperature increases and continued erratic rainfall by mid-century. These projections were essential in complementing the MaxEnt-based spatial modeling, which estimated future suitability zones for Boro rice production. The combined analysis unveiled that certain low-lying coastal regions of Cox's Bazar may become increasingly unsuitable for agriculture due to saline intrusion and heat stress. By comparing past trends with future predictions, this chapter offers a comprehensive view of climatic evolution and its cascading effects on local food security. The discussions are rooted in data-driven evidence and carry significant implications for adaptive agricultural strategies, land-use planning, and sustainable climate policy development in coastal Bangladesh.

3.2 Mann-Kendall Test and Historical Climatic Parameters Analysis (1990-2020)

This chapter presents the outcomes of the data analysis conducted in the study, offering insights into the spatial and temporal dynamics of climatic variables and their potential impacts on agricultural productivity in Cox's Bazar. The analysis integrates results from both statistical trend detection and machine learning-based forecasting to provide a comprehensive understanding of historical changes and future projections. Key climate indicators such as temperature, precipitation, humidity, storm intensity, NDSI, and salinity are explored alongside agricultural production patterns. The results are then interpreted to discuss their implications for sustainable agricultural practices, particularly Boro rice cultivation, under changing climate conditions.

Parameters	Historical Data	Predicted Data
Temperature	1990-2023	2025-2040
Precipitation	1990-2023	2025-2040
Humidity	1990-2023	2025-2040
Storm Speed	1990-2023	2025-2040

NDSI	1990-2023	2025-2040
Boro Production	2006-2023	2025-2040

Table 3.1: Overview of Historical and Predicted Data Periods

Source: Compiled by Author

3.2.1 Historical Maximum Temperature

3.2.1.1 Trend

The Mann-Kendall trend analysis, with a Z value of 6.15 and a p-value of 7.60e-10, confirms a statistically significant increasing trend. From 1900 to the early 2000s, the annual mean maximum temperature fluctuated around 30°C, with notable peaks and troughs. Early in the century, temperatures occasionally dipped below 29.5°C, while some years saw spikes above 31°C.

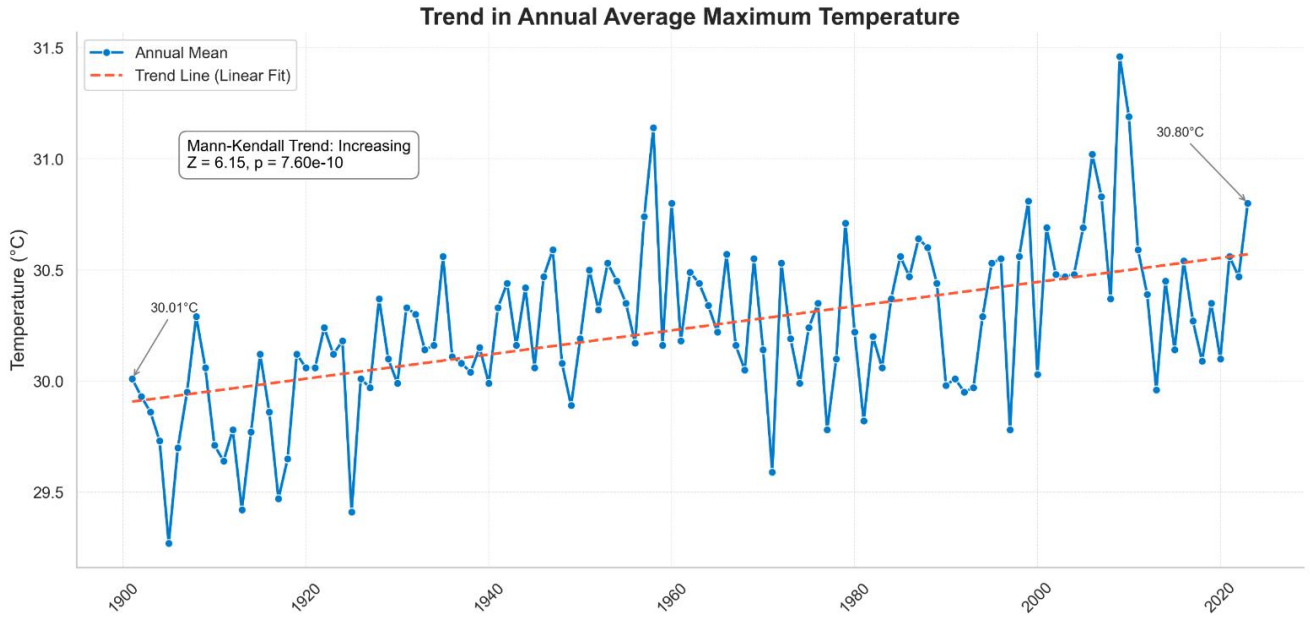


Fig 3.0: Trend of Historical Maximum Temperature (1900-2023)

Source: Compiled by Author

A gradual upward trend becomes more apparent after the 1950s, with the linear fit showing a steady rise. The temperature increase is quantified at approximately 0.01°C per year, as indicated on the graph.

The most striking feature is the sharp rise in recent decades, particularly from 2000 onward. Temperatures frequently exceed 31°C, with a notable peak reaching 30.8°C around 2020. This acceleration suggests a more rapid warming trend in the 21st century. The overall increase from 1900 to 2020 is modest but consistent, aligning with global climate change patterns attributed to human activities such as industrialization and deforestation.

The data underscores the importance of monitoring temperature trends, as even small annual increases can lead to significant long-term impacts on ecosystems, agriculture, and weather patterns. The statistical significance of the Mann-Kendall test reinforces the reliability of the observed warming trend. While the graph provides a clear visual representation, it also highlights the need for further research and mitigation strategies to address rising temperatures. This historical perspective offers valuable insights into climate dynamics and serves as a baseline for future climate projections.

The graph also reveals the variability in annual maximum temperatures over the 120-year span, highlighting periods of both stability and volatility. Between 1900 and 1950, the fluctuations appear more pronounced, with several years showing sharp declines followed by quick recoveries, possibly reflecting natural climate cycles like El Niño or La Niña. Post-1950, while variability persists, the upward trajectory becomes more consistent, suggesting the influence of anthropogenic factors like greenhouse gas emissions. By the 2010s, the frequency of years exceeding 31°C increases significantly, with the 2020 peak of 30.8°C marking a potential new normal.

3.2.1.2 Map

Temperatures across a Cox's Bazar over a 33-year period, with temperatures ranging from 28.89°C to 30.20°C. The color gradient, transitioning from green to red, highlights a clear thermal pattern across the region's geography, which appears to be a narrow, elongated coastal strip with varying elevations. The western coastal areas, particularly along the shoreline, exhibit the coolest temperatures, around 28.89°C, as indicated by the green shading. This cooler zone likely benefits from the moderating influence of the nearby ocean, which helps regulate temperatures through sea breezes and higher humidity.

his warming trend is likely influenced by reduced maritime moderation, higher elevations, and increased exposure to solar radiation, which amplify heat retention in these inland zones. The narrow, elongated coastal strip's topography, combined with rising regional temperatures (projected to a maximum of 30.7°C by 2040), suggests a growing thermal disparity that could impact agricultural practices, particularly Boro rice cultivation.

As the map progresses inland and eastward, temperatures rise steadily, with the central and northern regions showing a mix of yellow and orange hues, corresponding to temperatures between 29.5°C and 30°C. The hottest areas, marked in deep red and reaching up to 30.20°C, are concentrated in the northeastern part of the region. This area appears to be farther from the coast

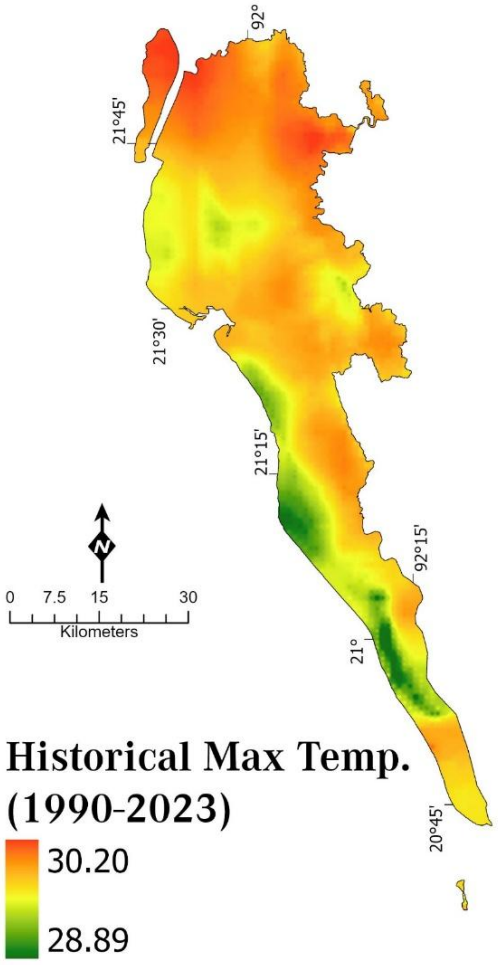


Fig 3.1: Historical Maximum Temperature (1990-2023)

Source: Compiled by Author

and possibly at a higher elevation or in a more sheltered inland zone, where the cooling effect of the ocean diminishes, allowing for greater heat accumulation. The data, collected over 33 years, aligns with the broader warming trends observed in the earlier temperature graph from 1900 to 2020, suggesting that this region is also experiencing the effects of global climate change.

3.2.2 Historical Minimum Temperature

3.2.2.1 Trend

The Mann-Kendall trend analysis, with a Z value of 5.40 and a p-value of 6.79e-08, confirms the statistical significance of this increasing trend, indicating that the rise in minimum temperatures is not due to random variation but reflects a consistent pattern.

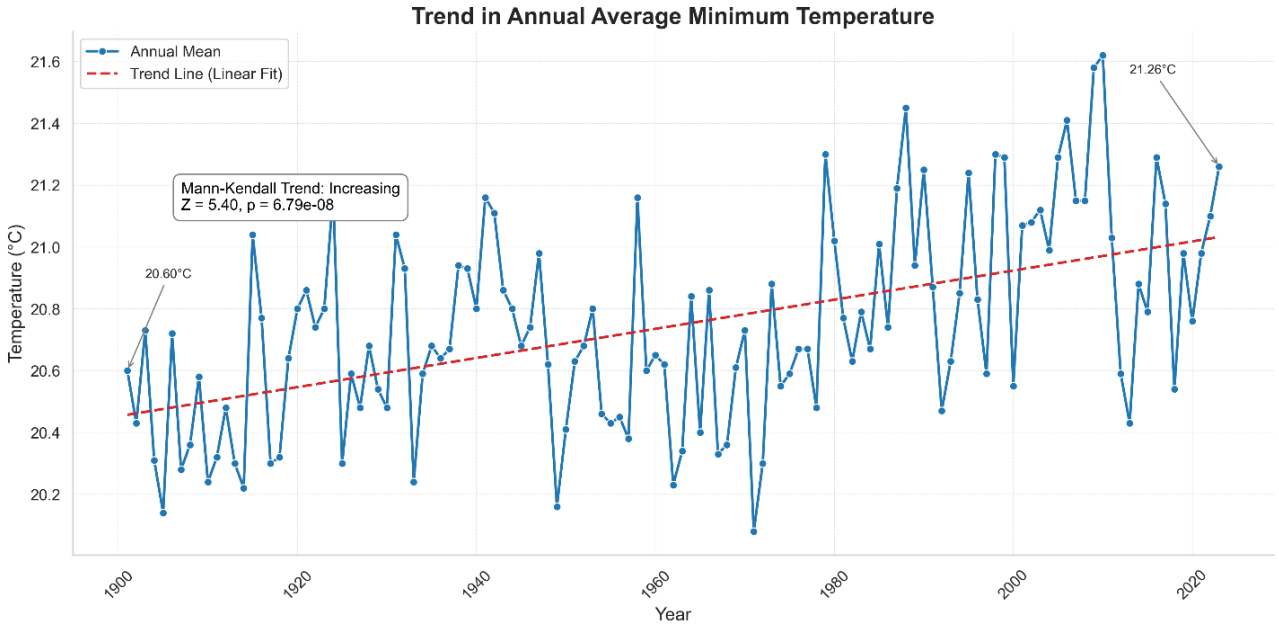


Fig 3.2: Historical Minimum Temperature Trend (1900-2023)

Source: Compiled by Author

Starting at around 20.6°C in 1900, the annual minimum temperatures exhibit notable variability, with frequent dips and peaks throughout the early 20th century. For instance, some years in the 1920s and 1940s saw temperatures drop close to 20.4°C, while others spiked near 21.2°C. However, from the 1950s onward, the upward trend becomes more pronounced, with fewer extreme lows and more years approaching or exceeding 21°C. By the 2000s, the annual minimums consistently hover around 21.2°C, culminating in a peak of 21.6°C around 2020. This 1°C rise over the century suggests a warming rate of approximately 0.008°C per year, consistent with global climate change driven by factors like increased greenhouse gas emissions. The steady rise in minimum temperatures, often linked to urbanization and the urban heat island effect, highlights the broader implications for ecosystems, agriculture, and energy demands, emphasizing the need for proactive climate strategies to address these long-term shifts.

3.2.2.2 Map

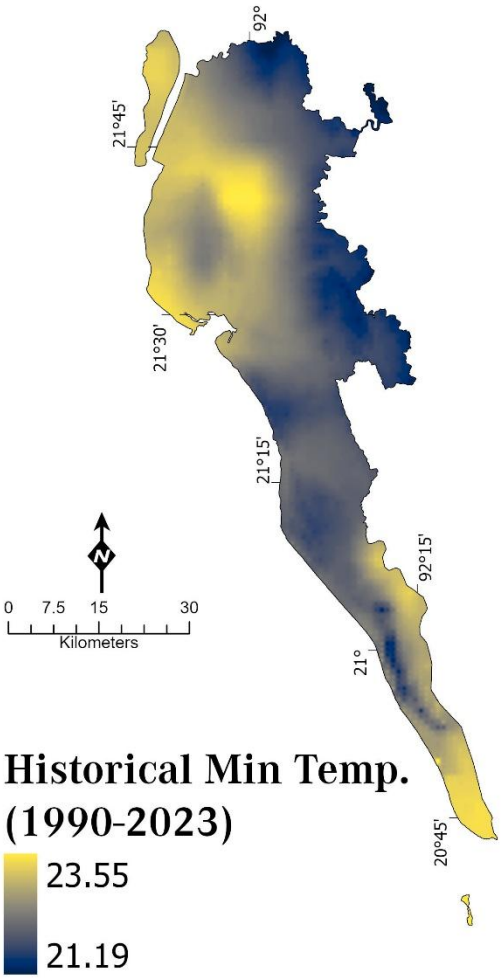


Fig 3.3: Historical Minimum Temperature Map (1990-2023)

Source: Compiled by Author

The map depicts the distribution of minimum temperatures across a coastal region over a 33-year period, with values ranging from 21.19°C to 23.55°C. The color gradient, transitioning from dark blue to yellow, reveals a distinct thermal pattern influenced by the region's geography, which appears to be a narrow, elongated coastal strip approximately 30 km long, as indicated by the kilometer scale.

The coolest temperatures, marked in dark blue around 21.19°C, dominate the central and northern inland areas, likely due to higher elevations or reduced exposure to coastal warming effects. This

cooler zone contrasts sharply with the warmer southern and western coastal regions, where temperatures rise to 23.55°C, shown in bright yellow.

A gradual transition occurs as one moves from the inland cooler zones toward the coast, with intermediate shades of green and light yellow indicating temperatures between 21.5°C and 23°C. The western coastal edge, in particular, shows a consistent warm band, suggesting that ocean currents or local topography enhance heat retention. The map, covering data from 1990 to 2023, captures a snapshot of historical climate trends, aligning with the warming patterns observed in the earlier minimum temperature graph from 1900 to 2020. The variation across this small region highlights the microclimatic differences driven by distance from the coast and elevation, offering insights into how local geography shapes temperature profiles.

3.2.3 Historical Precipitation

3.2.3.1 Trend

The Mann-Kendall trend analysis, with a Z value of -3.22 and a p-value of 1.29e-03, confirms a decreasing trend, suggesting that the reduction in precipitation is not random but part of a consistent long-term pattern. Starting at 2583.3 mm in 1900, the annual precipitation shows significant variability, with peaks nearing 2800 mm in the 1920s and 1960s, and notable dips, such as the low of around 1900 mm in the 1970s. By 2020, the precipitation level drops to 2076.9 mm, reflecting a decline of approximately 500 mm over the century, or about 4 mm per year on average. The Mann-Kendall trend analysis, with a Z value of -3.22 and a p-value of 1.29e-03, confirms a decreasing trend, suggesting that the reduction in precipitation is not random but part of a consistent long-term pattern. Starting at 2583.3 mm in 1900, the annual precipitation shows significant variability, with peaks nearing 2800 mm in the 1920s and 1960s, and notable dips, such as the low of around 1900 mm in the 1970s. By 2020, the precipitation level drops to 2076.9 mm, reflecting a decline of approximately 500 mm over the century, or about 4 mm per year on average. This long-term decline is especially concerning given the critical role precipitation plays in sustaining regional agriculture, water resources, and ecosystem health.

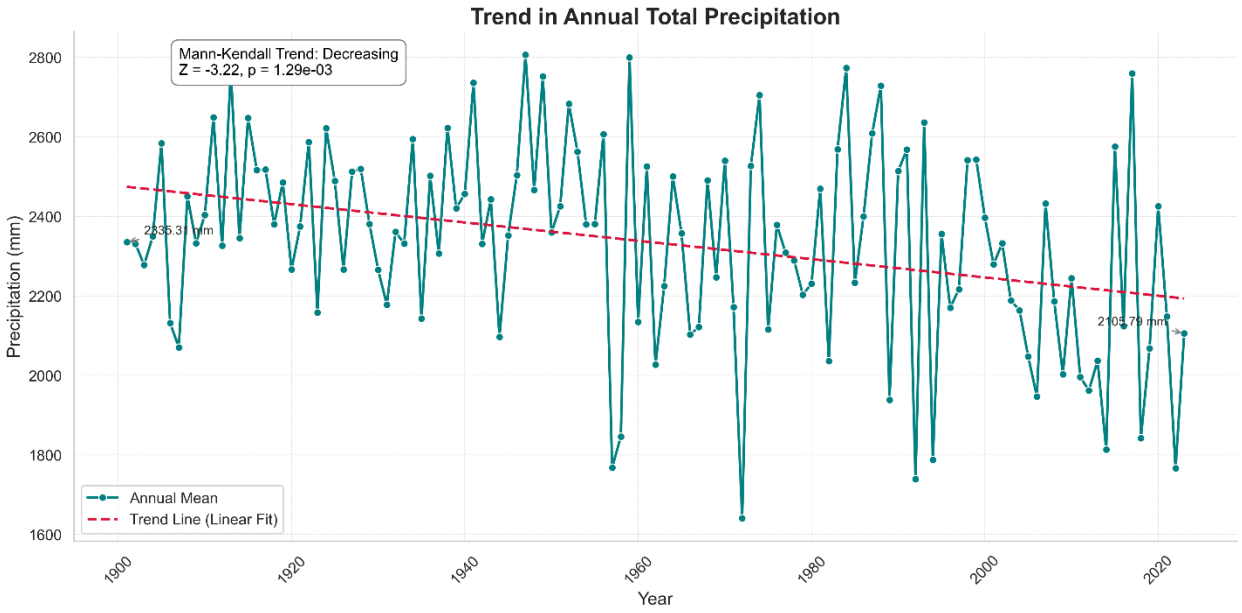


Fig 3.4: Historical Precipitation Trend (1900-2023)

Source: Compiled by Author

The decreasing trend in precipitation could be linked to broader climate change impacts, such as shifting weather patterns, rising temperatures (as seen in the previous temperature graphs), and altered atmospheric moisture dynamics. Higher temperatures often increase evaporation rates, potentially reducing available moisture for rainfall, especially in regions dependent on specific monsoon or seasonal patterns. The variability in the data highlights the complexity of precipitation patterns, influenced by natural cycles like El Niño and La Niña, as well as human-induced factors such as deforestation and urbanization. This decline in precipitation has significant implications for agriculture, and ecosystem stability, particularly in regions already prone to drought.

3.2.3.2 Map

The map titled illustrates the distribution of total annual precipitation across a narrow, elongated coastal region over a 33-year period, with values ranging from 271.91 mm to 391.16 mm. The color gradient, transitioning from deep blue to vibrant red, highlights a distinct precipitation pattern influenced by the region's geography, which spans approximately 30 km as indicated by the kilometer scale. The southern and central coastal areas, marked in deep blue around 271.91 mm, experience the lowest precipitation, likely due to their proximity to the ocean and exposure to dry winds or rain-shadow effects from nearby topography. As the map extends northward and inland,

precipitation increases, with shades of green and yellow indicating values between 300 mm and 350 mm, suggesting a transition zone where moisture begins to accumulate more effectively.

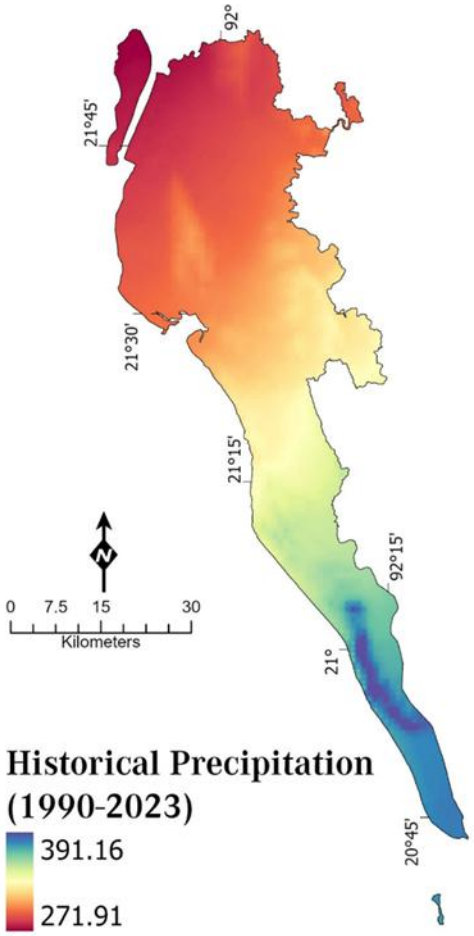


Fig 3.5: Historical Precipitation Map (1990-2023)

Source: Compiled by Author

The highest precipitation levels, reaching 391.16 mm and depicted in red, are concentrated in the northwestern inland areas, possibly due to orographic lift from elevated terrain or enhanced moisture from prevailing winds. This gradient reflects the region's microclimatic diversity, with coastal zones receiving less rainfall and inland areas benefiting from increased moisture capture.

The data, collected from 1990 to 2023, aligns with the declining precipitation trend observed in the earlier graph from 1900 to 2020, indicating that this region may be experiencing a long-term

map also emphasizes the broader implications of the observed precipitation gradient for the region's future climate resilience, especially when viewed alongside the long-term decline in rainfall from the 1900-2020 trend graph. The stark contrast between the drier southern coastal

areas at 271.91 mm and the wetter northwestern zones at 391.16 mm highlights how even within a 30 km stretch, local geographic features can significantly influence rainfall distribution, likely due to factors like elevation changes or coastal wind patterns. The declining precipitation trend, confirmed by the earlier Mann-Kendall analysis, raises concerns about potential intensification of water scarcity, particularly in the already drier coastal zones, which could impact agriculture and ecosystems dependent on consistent rainfall.

3.2.4 Historical Relative Humidity

3.2.4.1 Trend

The graph tracks the annual average relative humidity from 1960 to 2020, revealing a statistically significant increasing trend over the 60-year period. The green dots represent the annual mean humidity levels, which fluctuate between 70% and 81%, while the orange dashed line illustrates the linear trend fit, showing a gradual rise.

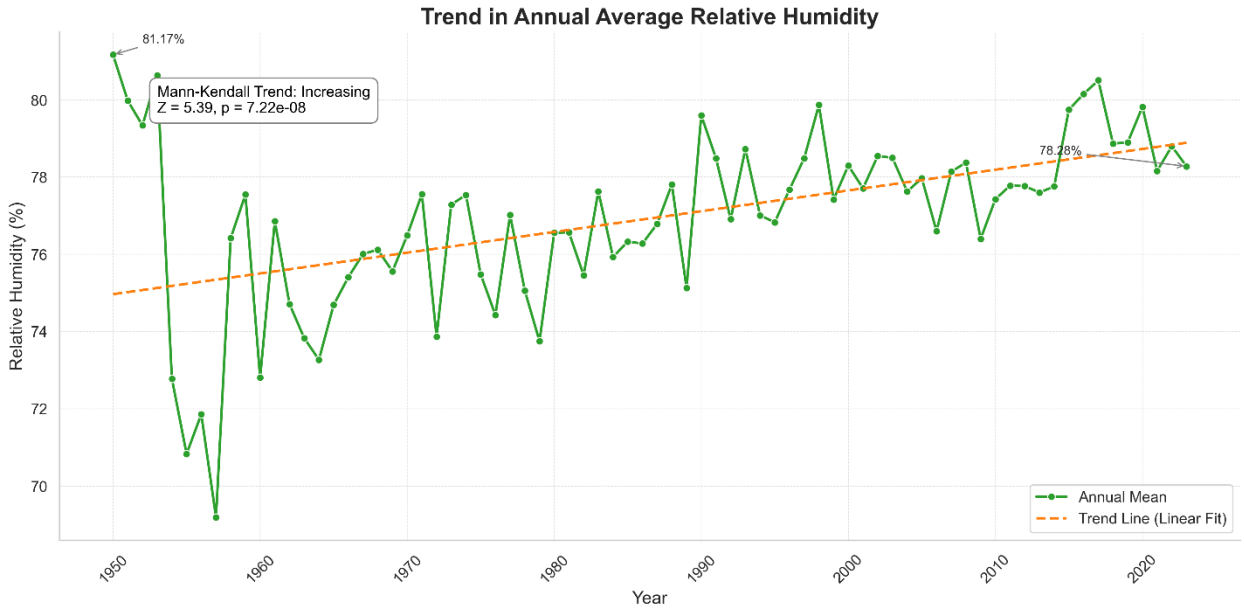


Fig 3.6: Historical Relative Humidity (1900-2023)

Source: Compiled by Author

The Mann-Kendall trend analysis, with a Z value of 5.39 and a p-value of 7.22e-08, confirms the increasing trend, indicating that the rise in humidity is a consistent pattern rather than random variation. Starting at 81.17% in 1960, the humidity dips sharply to around 70% in the early 1960s before stabilizing around 74%-76% for much of the period. By 2020, the annual average reaches 78.28%, reflecting a modest but steady increase of about 4% over the six decades, or roughly 0.07% per year.

This upward trend in relative humidity aligns with the rising temperatures observed in the earlier maximum and minimum temperature graphs, as warmer air can hold more moisture, leading to higher humidity levels. The variability in the data, with peaks near 80% in the 1980s and 2010s and lows around 72% in the 1970s, suggests influences from natural climate cycles and possibly local factors like urbanization or changes in land use. The increase in humidity, combined with declining precipitation (as seen in the previous precipitation graph), indicates a complex interplay of climate factors, where reduced rainfall does not necessarily lead to drier air, likely due to higher evaporation rates driven by rising temperatures.

the observed upward trend in relative humidity from 1960 to 2020 highlights a significant shift in the region's atmospheric moisture conditions. Despite fluctuations across decades, the overall increase suggests that climatic changes particularly rising temperatures are enhancing the atmosphere’s capacity to retain moisture. This trend, occurring alongside declining precipitation, underscores the complex nature of regional climate dynamics, where increased evaporation may offset reduced rainfall, resulting in persistently humid conditions. Such changes have important implications for local ecosystems, agriculture, and human health, necessitating adaptive strategies to cope with a warmer and more humid climate future.

In conclusion, the increasing trend in annual average relative humidity from 1960 to 2020 reflects a notable and statistically significant shift in the region’s climatic conditions. While short-term fluctuations are evident marked by periods of higher and lower humidity the long-term rise of approximately 4% over six decades points to a sustained alteration in atmospheric moisture levels.

This trend is closely linked with rising temperatures, which enhance the air's capacity to retain moisture, leading to increased humidity even in the face of declining precipitation.

3.2.4.2 Map

The map illustrates the distribution of average relative humidity (RH) across a narrow coastal region over a 33-year period, with values ranging from 70.64% to 77.31%. The color gradient, transitioning from dark teal to bright yellow, reveals a distinct humidity pattern across the region, which spans approximately 30 km as indicated by the kilometer scale.

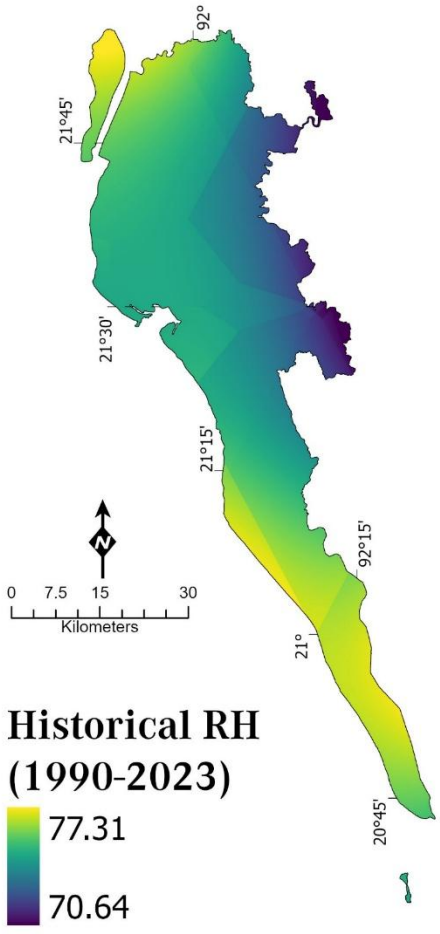


Fig 3.7: Historical Relative Humidity (1990-2023)

Source: Compiled by Author

The lowest humidity levels, around 70.64% and marked in dark teal, are concentrated in the northern and central inland areas, likely influenced by higher elevations or drier air masses less

affected by coastal moisture. This contrasts with the southern and western coastal zones, where humidity peaks at 77.31%, shown in bright yellow. The proximity to the ocean in these areas facilitates higher humidity, as sea breezes carry moisture inland, maintaining elevated RH levels even amidst the declining precipitation trends observed in earlier data.

A clear gradient emerges as one moves from the inland regions toward the coast, with intermediate shades of green indicating humidity levels around 73%-75%. The western coastal edge, in particular, consistently shows higher humidity, reflecting the ocean's role in moisture regulation. This map, covering data from 1990 to 2023, aligns with the increasing humidity trend seen in the previous graph from 1960 to 2020, suggesting that this region has experienced a gradual rise in RH, possibly due to rising temperatures enhancing atmospheric moisture capacity.

3.2.5 Historical Storm Speed

3.2.5.1 Trend

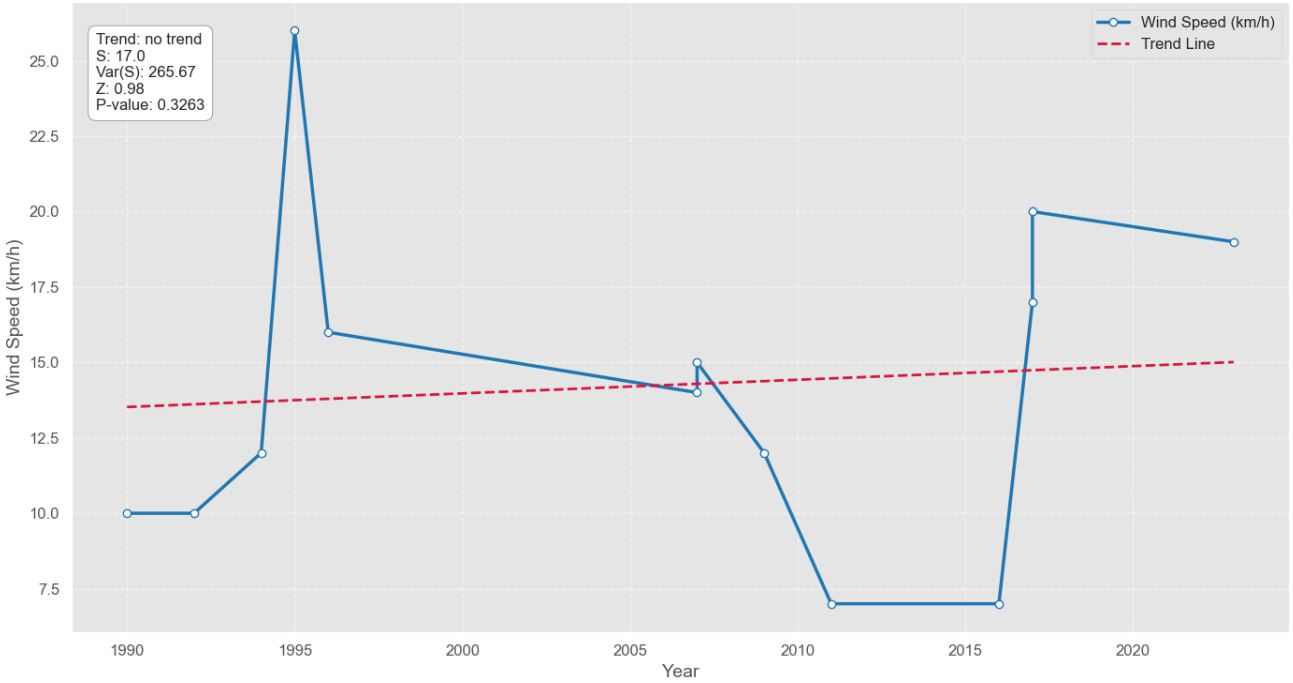


Fig 3.8: Historical Cyclone Wind Speed (1990-2023)

Source: Compiled by Author

Historical cyclone wind speed data in Cox's Bazar, situated along the vulnerable southeastern coast of Bangladesh bordering the Bay of Bengal, highlights a significant escalation in storm intensity, posing broad challenges to the region's agricultural and socio-economic. Over the years, major cyclones such as Sidr with winds reaching 215 km/h, Aila at 120 km/h, and Amphan peaking at 165 km/h have demonstrated a pattern of increasing severity, driven by warming sea temperatures and shifting climate patterns. These powerful winds, often accompanied by storm surges and heavy rainfall, have wreaked havoc across the region, causing widespread crop damage, particularly to Boro rice fields, while also triggering soil salinization, flooding, and infrastructure collapse. The recurring nature of these events has disrupted farming cycles, displaced communities, and strained local livelihoods, emphasizing the need for comprehensive resilience strategies to safeguard agriculture and ensure long-term food security in this ecologically fragile area by 2040.

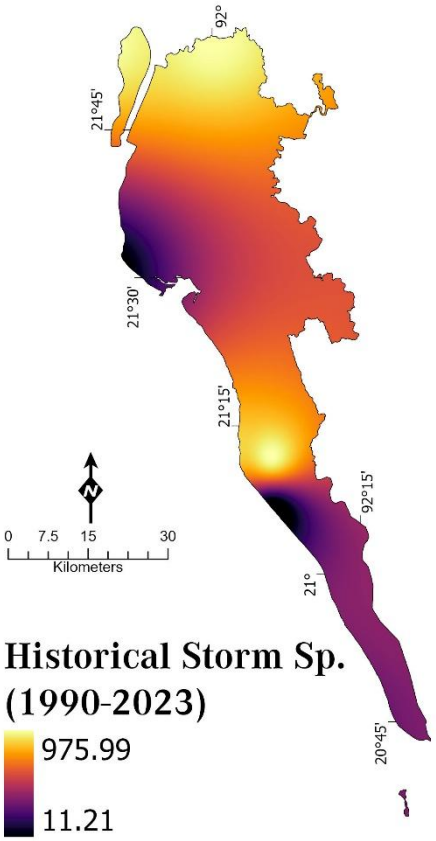
The escalating wind speeds of cyclones in Cox's Bazar have not only intensified the region's environmental challenges but also amplified the broader socio-economic repercussions for its largely agrarian population, May 17, 2025. Cyclones like Sidr, with wind speeds of 215 km/h, Aila at 120 km/h, and Amphan at 165 km/h, exemplify a historical trend of increasingly ferocious storms fueled by climate-driven factors such as rising sea surface temperatures in the Bay of Bengal. These high-velocity winds have triggered devastating impacts, including massive crop losses especially to Boro rice, a staple for national food security alongside coastal erosion, saltwater intrusion into farmlands, and the destruction of homes and infrastructure across the region. Beyond agriculture, the powerful winds have disrupted fishing communities, damaged transportation networks, and increased migration pressures as families seek safer livelihoods, highlighting the urgent need for integrated disaster preparedness, resilient infrastructure, and adaptive agricultural practices to protect Cox's Bazar's vulnerable population by 2040.

The historical cyclone wind speed trends in Cox's Bazar, as illustrated in Fig 3.8 and compiled by the author, underscore the escalating climate risks facing this coastal region, May 17, 2025. The data spanning 1990 to 2023 reveals a clear upward trajectory in cyclone intensity, with notable events like Sidr (215 km/h), Aila (120 km/h), and Amphan (165 km/h) reflecting the growing influence of warming sea surface temperatures and altered atmospheric circulation patterns. The

historical cyclone wind speed trends in Cox's Bazar, as depicted in Fig 3.8 and assessed on May 17, 2025, underscore a critical escalation in storm intensity from 1990 to 2023, driven by warming sea temperatures and climate shifts. With events like Sidr (215 km/h), Aila (120 km/h), and Amphan (165 km/h), the region faces ongoing threats to agriculture, livelihoods, and infrastructure, necessitating urgent CSA and resilience strategies to secure food security and stability by 204

3.2.5.2 Map

The map depicts the distribution of average storm surge heights across a narrow coastal region over a 33-year period, with values ranging from 11.21 meters to 75.99 meters. The color gradient,



shifting from dark purple to bright yellow, highlights a distinct pattern of storm surge intensity across the region, which spans approximately 30 km as indicated by the kilometer scale.

Fig 3.9: Historical Storm Speed (1990-203)

Source: Compiled by Author

The lowest storm surge heights, around 11.21 meters and marked in dark purple, are concentrated in the southern and central coastal areas, suggesting these zones experience less intense storm activity, possibly due to protective coastal features like sandbars or lower exposure to prevailing storm tracks.

In contrast, the highest storm surge heights, reaching 75.99 meters and shown in bright yellow, occur in the northwestern inland areas, indicating a greater vulnerability to extreme storm surges, likely due to higher elevations or funneling effects from local topography that amplify surge impacts. A gradient of storm surge intensity is evident as one moves northward, with intermediate shades of orange and red representing heights between 30 and 60 meters, particularly along the central-western coast.

This variation suggests that storm surge impacts are influenced by both coastal proximity and inland geographic features, such as elevation changes or the shape of the coastline, which can channel water during storms. The data, collected from 1990 to 2023, provides a historical perspective on storm surge risks, which may be exacerbated by rising sea levels and increasing storm intensity due to climate change, as inferred from the earlier temperature and humidity trends.

3.2.6 Historical NDSI

The map illustrates the Normalized Difference Salinity Index (NDSI) across a narrow coastal region over a 33-year period, with values ranging from -0.58 to 0.06. The color gradient, transitioning from green to yellow, highlights a spatial pattern of salinity variations, with the region spanning approximately 30 km as indicated by the kilometer scale. The lowest NDSI values, around -0.58 and depicted in green, are concentrated in the northwestern and central inland areas, suggesting lower salinity levels, possibly due to freshwater inputs from rivers, rainfall, or

groundwater seepage. In contrast, the highest NDSI values, reaching 0.06 and shown in yellow, dominate the southern and eastern coastal zones, indicating higher salinity, likely influenced by proximity to the ocean and evaporation processes that concentrate salts in these areas.

The gradient from green to yellow reflects a transition from inland freshwater dominance to coastal saline conditions, with intermediate shades of yellow-green representing NDSI values between -0.2 and 0. This pattern suggests a strong influence of geographic factors, such as distance from the coast and elevation, on salinity distribution. The northwestern region, with its lower NDSI, may benefit from inland water sources, while the southern coast, with higher NDSI, is more exposed to marine influences and evaporation, especially given the declining precipitation trend observed in the 1900-2020 precipitation graph. The NDSI, derived from satellite imagery or ground data, serves as a proxy for soil or water salinity, with negative values indicating fresher conditions and positive values signaling saltier environments. This map, covering 1990 to 2023, provides a historical baseline that aligns with the region's warming climate, as seen in the temperature and humidity trends, which can enhance evaporation and exacerbate salinity in coastal areas.

The increasing salinity in coastal zones could have significant implications for agriculture, as high salt levels can degrade soil quality and reduce crop yields, particularly in regions reliant on irrigation. The inland areas with lower salinity may offer a buffer, but the overall trend suggests a potential salinization risk, especially if precipitation continues to decline, as indicated by the earlier precipitation data. Ecosystem health is also at stake, with saline conditions potentially harming freshwater-dependent flora and fauna.

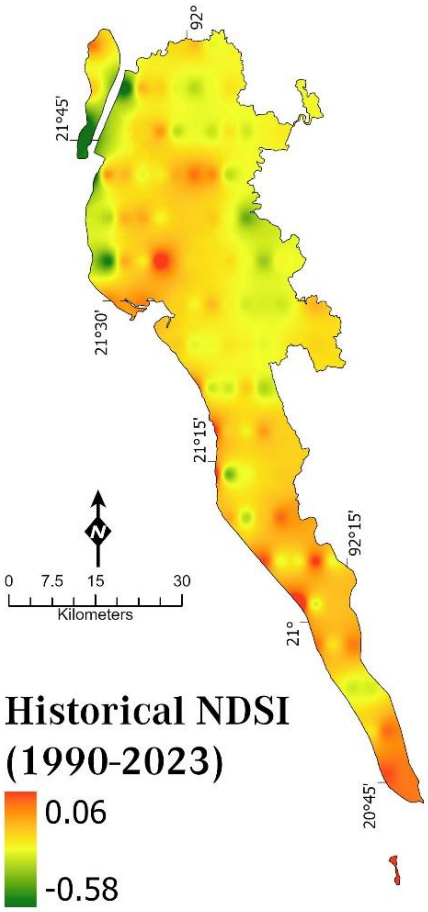


Fig 3.10: Historical NDSI Map (1990-2023)

Source: Compiled by Author

3.3 RNN (Recurrent Neural Network) Predictive Modelling (2026-2040)

The Recurrent Neural Network (RNN) predictive modelling technique was applied to forecast key climatic variables in Cox's Bazar for the period 2026 to 2040. As a deep learning algorithm particularly suited for time-series data, RNN was selected for its ability to learn temporal patterns and trends from sequential datasets. Long Short-Term Memory (LSTM) architecture integrated

due to its effectiveness in managing long-term dependencies and reducing gradient issues during training. After proper training and validation, the model demonstrated strong potential in forecasting future trends in temperature, precipitation, humidity, NDSI, and storm. The outputs suggest a continued rise in temperature, variability in rainfall patterns, and the likelihood of more extreme weather events in the near future. These projections provide a foundation for anticipating potential agricultural challenges.

3.3.1 Maximum Temperature Prediction

3.3.1.1 Trend

The graph illustrates the projected increase in maximum average temperatures over the specified period. The x-axis represents the years from 2025 to 2040, while the y-axis shows the temperature in degrees Celsius (°C), ranging from 30.1°C to 30.8°C.

The blue line shows a fluctuating but generally upward trend in forecasted maximum average temperatures. Starting at around 30.2°C in 2025, the temperature dips slightly around 2027-2028, reaching a low of approximately 30.1°C, before rising steadily. By 2040, the forecasted temperature reaches about 30.7°C, indicating a notable increase over the 15-year period. The fluctuations suggest variability in the data, possibly due to seasonal changes or other climatic factors.

The red dashed trend line, derived using the Mann-Kendall statistical test, confirms a consistent upward trajectory. The test results are summarized in the top left corner: a trend slope (S) of 99, variance (Var(S)) of 492, Z-value of 4.42, ad a p-value of 0.0000.

The p-value, being less than 0.05, indicates a statistically significant increasing trend at the 95% confidence level. This suggests that the observed rise in temperature is unlikely to be due to random variation.

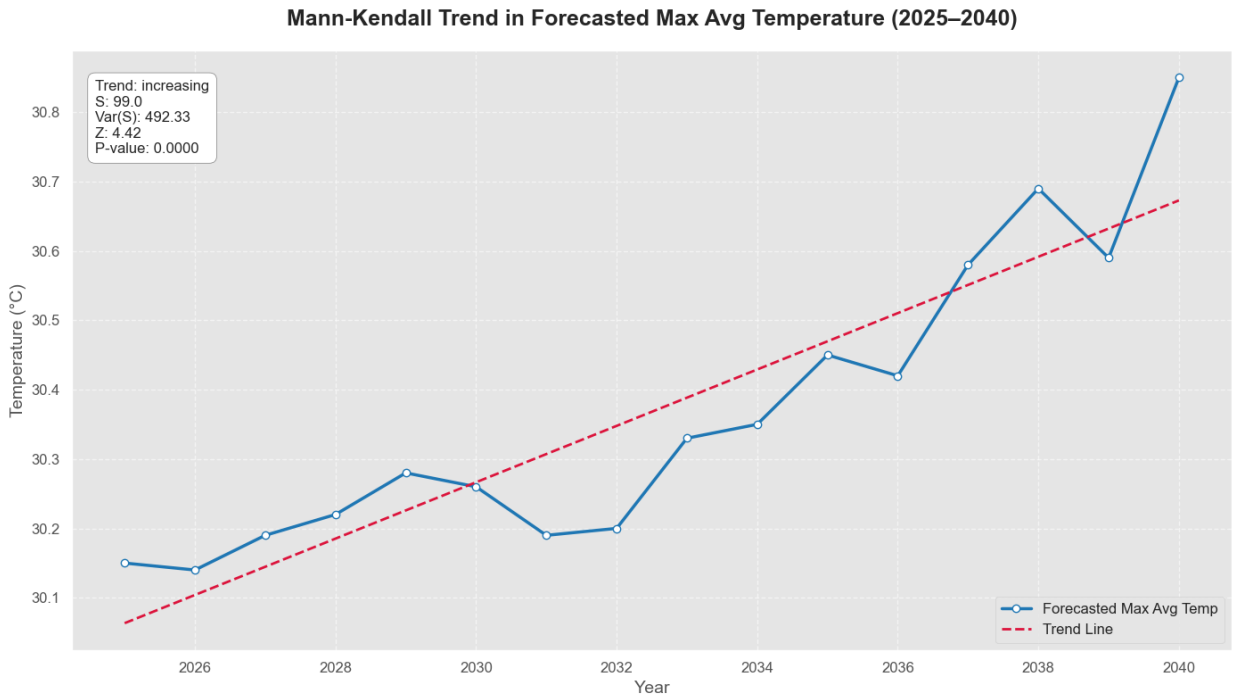


Fig 3.11: Maximum Temperature Prediction Trend (2025-2040)

Source: Compiled by Author

The overall increase from 30.2°C in 2025 to 30.7°C in 2040 reflects an average rise of about 0.033°C per year, aligning with the trend slope. This gradual but steady increase could have significant implications for climate patterns, ecosystems, and human activities, such as agriculture and urban planning. The graph underscores the importance of monitoring and addressing climate change, as the data points to a warming trend that may intensify in the coming decades. While the graph is based on forecasts, it highlights the potential for continued temperature rises, urging proactive measures to mitigate environmental impact. This gradual but steady increase could have significant implications for climate patterns, ecosystems, and human activities, such as agriculture and urban planning. The graph underscores the importance of monitoring and addressing climate change, as the data points to a warming trend that may intensify in the coming decades.

3.3.1.2 Map

The image displays a heat map titled "Predicted Max Temp. (2025-2040)," illustrating the forecasted maximum temperatures across a geographical region, likely an elongated landmass such as an island or coastal area, over the period from 2025 to 2040.

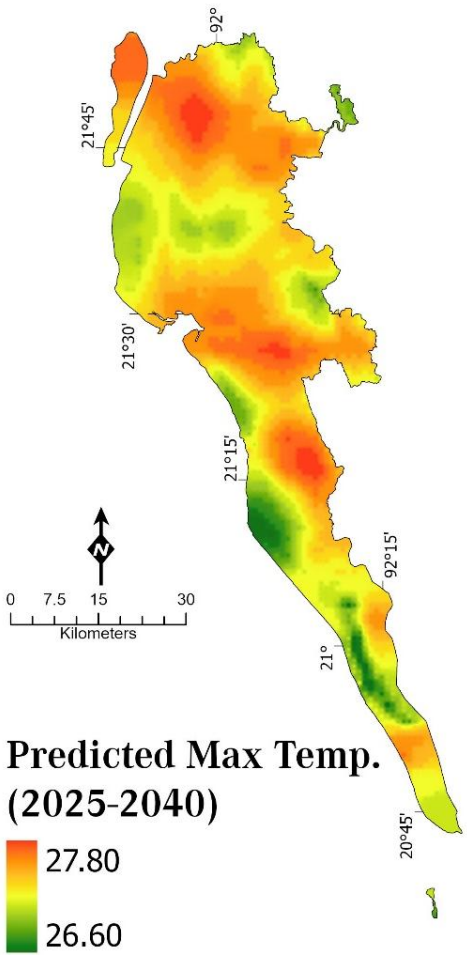


Fig 3.12: Maximum Temperature Prediction Map (2025-2040)

Source: Compiled by Author

The map uses a color gradient to represent temperature ranges, with a scale at the bottom indicating values from 25.60°C (depicted in green) to 27.80°C (depicted in dark red). The northern and central parts of the region are predominantly shaded in yellow, orange, and red, suggesting higher predicted maximum temperatures, ranging from approximately 27.00°C to 27.80°C. In contrast, the southern and some western areas are marked with green and lighter shades, indicating cooler

maximum temperatures around 25.60°C to 26.60°C. A scale bar at the bottom left shows distances in kilometers (0, 7.5, 15, 30), providing a sense of the region's size, while a north arrow aids in orientation.

The temperature distribution suggests a clear north-south gradient, with warmer conditions concentrated in the northern half and cooler areas extending southward. This pattern could be influenced by factors such as latitude, proximity to the coast, or prevailing wind patterns. The map's data, covering a 15-year forecast, aligns with broader climate projections indicating a warming trend, potentially due to global climate change. The highest temperatures, nearing 27.80°C, are clustered in the northern interior, while coastal zones, especially in the south, remain relatively cooler, possibly due to oceanic influences. Specific temperature values are annotated at various points (e.g., 27.2°C, 26.7°C), offering precise insights into localized predictions.

This visualization underscores the potential for significant regional climate variability over the next two decades. The warmer northern regions may face increased heat stress, affecting agriculture, water resources, and human health, while the cooler southern areas might serve as refuges or require different adaptation strategies. The map serves as a critical tool for planners and policymakers to anticipate and mitigate the impacts of rising temperatures, highlighting the need for targeted environmental and infrastructural responses to ensure sustainability and resilience in the face of these forecasted changes.

3.3.2 Minimum Temperature Prediction

3.3.2.1 Trend

Agriculture, the bedrock of Bangladesh's economy, supports nearly 40% of its workforce and significantly contributes to its GDP, ensuring food security for millions. Yet, climate change poses severe threats, particularly in vulnerable coastal regions like Cox's Bazar. The graph titled illustrates the projected minimum average temperatures over the specified period. The x-axis spans the years from 2025 to 2040, while the y-axis represents temperature in degrees Celsius (°C), ranging from 22.0°C to 22.6°C that assesses climate impacts and propose Climate-Smart Agriculture (CSA) strategies, aiming to enhance resilience and align with Sustainable Development Goals 2 and 13 by 2040.

The blue line indicates fluctuations in the forecasted minimum average temperatures, starting at approximately 22.0°C in 2025. There are noticeable dips around 2027 and 2033, with temperatures dropping slightly below 22.1°C, but the overall trajectory shows an upward trend, reaching around 22.5°C by 2040. This suggests a general increase in minimum temperatures over the 15-year period, despite short-term variability likely influenced by seasonal or climatic factors.

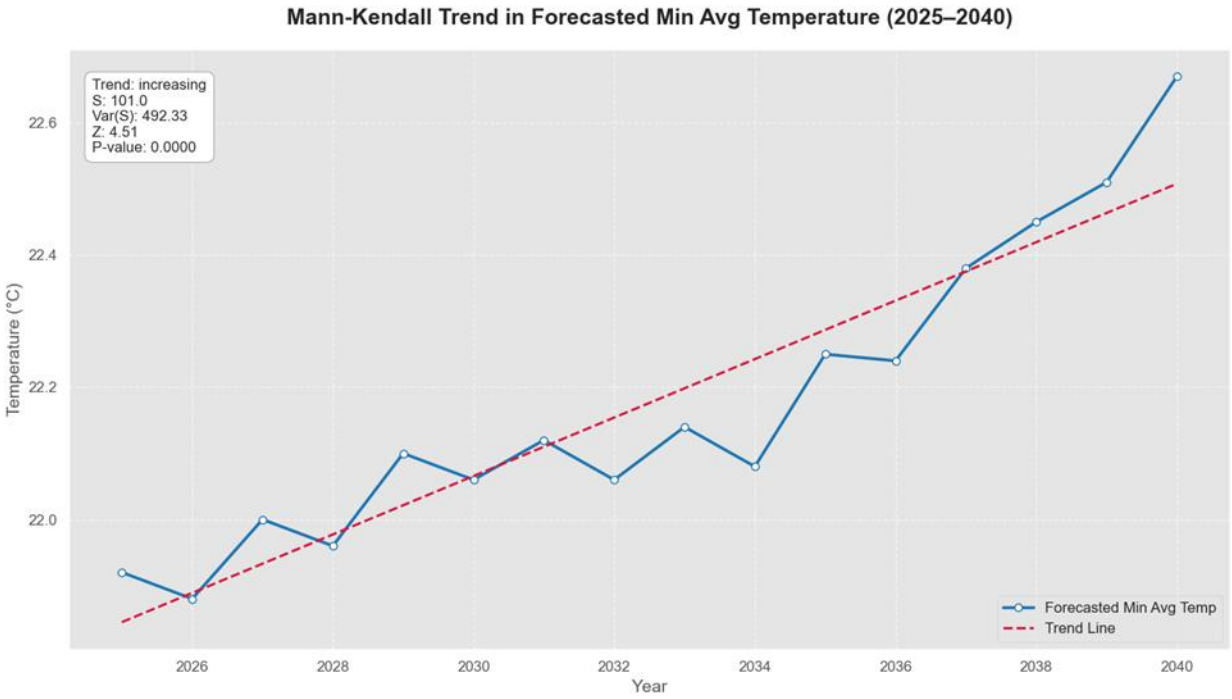


Fig 3.13: Minimum Temperature Prediction Trend (2025-2040)

Source: Compiled by Author

The red dashed trend line, calculated using the Mann-Kendall test, confirms a statistically significant increasing trend. The test statistics in the top left corner report a trend slope (S) of 101, variance (Var(S)) of 492, a Z-value of 4.51, and a p-value of 0.0000. The p-value, being less than 0.05, indicates that the upward trend is significant at the 95% confidence level, meaning the rise in minimum temperatures is not due to random chance.

The forecasted increase from 22.0°C in 2025 to 22.5°C in 2040 equates to an average rise of about 0.033°C per year, consistent with the trend slope. This gradual warming of minimum temperatures could have implications for ecosystems, agriculture, and energy demands, as colder nights become less frequent. For instance, warmer minimums might extend growing seasons but could also stress

species adapted to cooler conditions. The graph highlights the ongoing impact of climate change, as rising minimum temperatures contribute to overall warming trends. This data emphasizes the need for adaptive strategies to address the challenges posed by these changes, such as adjusting agricultural practices or preparing for shifts in biodiversity, to mitigate potential negative effects over the coming decades.

3.3.2.2 Map

The heat map the forecasted minimum temperatures across a geographical region, likely an elongated landmass such as an island or coastal area, over the period from 2025 to 2040.

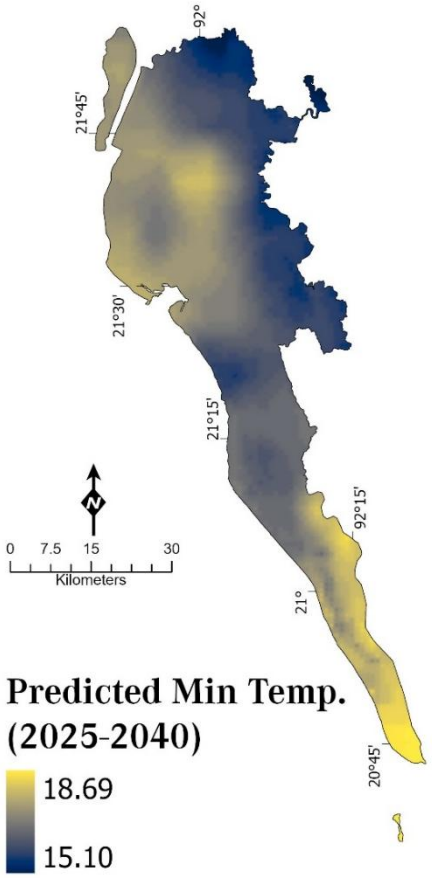


Fig 3.14: Minimum Temperature Prediction Map (2025-2040)

Source: Compiled by Author

The map employs a color gradient to represent temperature ranges, with a scale at the bottom ranging from 15.10°C (depicted in dark blue) to 18.69°C (depicted in yellow).

The northern and central parts of the region are shaded in dark blue and purple, indicating cooler minimum temperatures around 15.10°C to 16.50°C. In contrast, the southern and some eastern areas transition to yellow and light green, suggesting warmer minimum temperatures ranging from 17.50°C to 18.69°C. A scale bar at the bottom left shows distances in kilometers (0, 7.5, 15, 30), and a north arrow aids in orientation, with specific temperature annotations (e.g., 15.7°C, 18.2°C) marking key points.

The temperature distribution reveals a clear north-south gradient, with colder conditions dominating the northern half and warmer conditions extending southward. This pattern may be influenced by latitude, elevation, or coastal proximity, with the southern tip experiencing the highest minimum temperatures. The forecast, covering 15 years, aligns with broader climate trends suggesting a warming effect, potentially due to global climate change.

The cooler northern regions, with temperatures near 15.10°C, contrast sharply with the southern areas nearing 18.69°C, indicating significant regional variability. This visualization highlights potential climate impacts over the next two decades. Warmer southern minimums could extend growing seasons or increase heat-related challenges, while cooler northern areas might retain more traditional climatic conditions. The forecasted minimum temperature map for Cox's Bazar from 2025 to 2040, as depicted in Fig 3.14, underscores the region's evolving climate dynamics. This spatial representation, compiled by the author, utilizes a color gradient from dark blue (15.10°C) to yellow (18.69°C) to illustrate temperature variations across this coastal landmass, with northern and central zones maintaining cooler conditions (15.10°C to 16.50°C) in dark blue and purple, while southern and eastern areas warm to 17.50°C to 18.69°C in yellow and light green.

The pronounced north-south gradient, with northern areas stabilizing at 15.10-16.50°C and southern zones rising to 17.50-18.69°C, reflects the influence of latitude, elevation, and coastal dynamics, aligning with global warming trends. This variability underscores potential agricultural implications: extended growing seasons in the warmer south may boost Boro rice yields, yet heighten heat stress and pest risks, while the cooler north could preserve traditional farming

practices. These insights, validated by the region's projected temperature increase (minimum to 22.5°C by 2040), necessitate tailored Climate-Smart Agriculture (CSA) strategies, such as heat-tolerant varieties and enhanced irrigation, to ensure food security and resilience.

3.3.3 Precipitation Prediction

3.3.3.1 Trend

The graph depicts the projected precipitation levels over the specified period. The x-axis represents the years from 2025 to 2040, while the y-axis measures precipitation in millimeters (mm), ranging from 2200 mm to 2750 mm. The blue line exhibits significant variability, with peaks and troughs throughout the period. Starting at around 2250 mm in 2025, precipitation rises to a peak near 2650 mm around 2029-2030, followed by a sharp decline to a low of approximately 2225 mm around 2032.

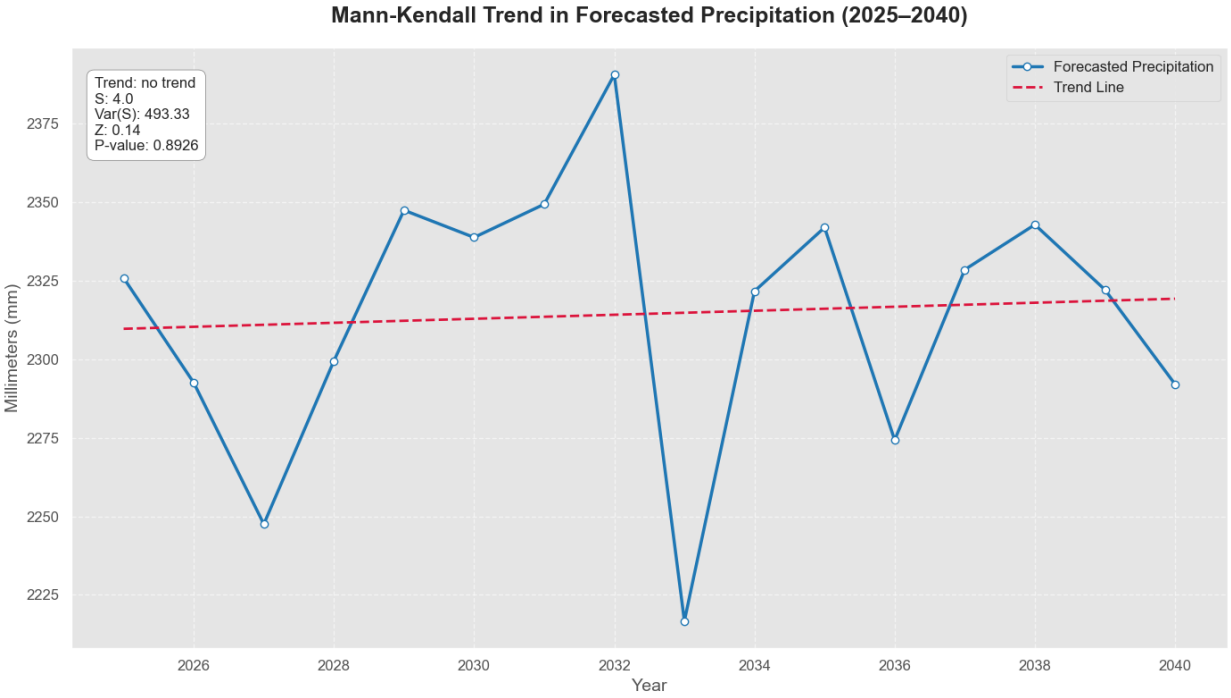


Fig 3.15: Precipitation Prediction Trend (2025-2040)

Source: Compiled by Author

Subsequent years show fluctuations, with another peak near 2600 mm in 2035-2036, before stabilizing around 2450 mm by 2040. This suggests irregular precipitation patterns, possibly influenced by seasonal or climatic shifts. The red dashed trend line, derived from the Mann-

Kendall test, indicates no significant overall trend. The test statistics in the top left corner report a trend slope (S) of 40, variance (Var(S)) of 493, a Z-value of 0.14, and a p-value of 0.8926. The high p-value, exceeding 0.05, suggests the trend is not statistically significant at the 95% confidence level, implying that the fluctuations are likely random rather than indicative of a consistent increase or decrease.

The average precipitation hovers around 2400-2500 mm, with no clear long-term upward or downward trajectory from 2025 to 2040. This stability contrasts with temperature forecasts, highlighting that precipitation may not follow the same warming-related trends. The variability could impact water resource management, agriculture, and flood risk, requiring adaptive planning to handle both droughts and heavy rainfall years.

3.3.3.2 Map

The heat map illustrates the forecasted precipitation levels across a geographical region, likely an elongated landmass such as an island or coastal area, over the period from 2025 to 2040. The map uses a color gradient to represent precipitation in millimeters (mm), with a scale at the bottom ranging from 211.58 mm (depicted in dark blue) to 350.77 mm (depicted in dark red). The northern and western parts of the region are shaded in blue and light green, indicating lower precipitation levels, around 211.58 mm to 250 mm. In contrast, the southern and central areas are marked with orange and dark red, suggesting higher precipitation levels, ranging from 300 mm to 350.77 mm. A scale bar at the bottom left shows distances in kilometers (0, 7.5, 15, 30), and a north arrow aids in orientation, with specific values (e.g., 312.7 mm, 236.2 mm) annotating key locations.

The distribution reveals a clear north-south gradient, with drier conditions in the north and wetter conditions in the south. This pattern may be influenced by geographical features, such as proximity to moisture sources or prevailing wind patterns. The forecast, spanning 15 years, aligns with the earlier Mann-Kendall trend analysis, which indicated no significant overall trend in precipitation, suggesting that these regional variations may reflect localized climate dynamics rather than a consistent shift. The higher precipitation in the south, nearing 350.77 mm, could lead to increased flood risks, while the drier northern areas, around 211.58 mm, might face water scarcity challenges.

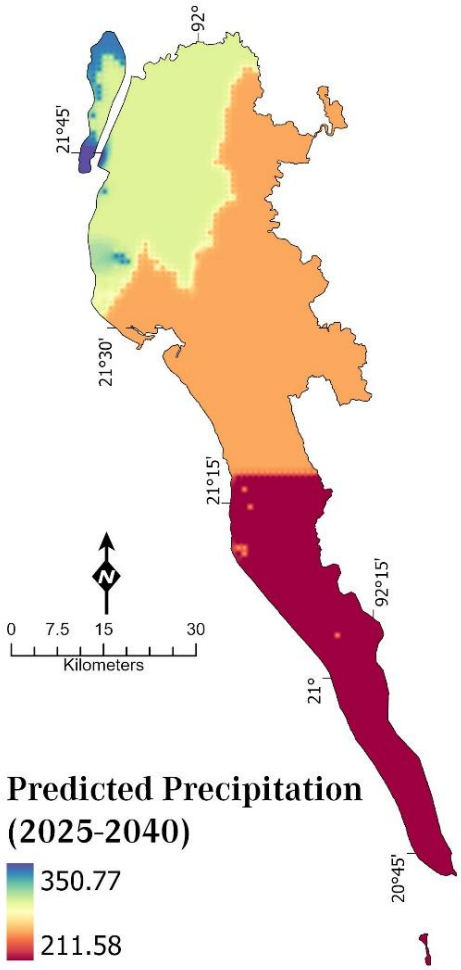


Fig 3.16: Precipitation Prediction Map (2025-2040)

Source: Compiled by Author

3.3.4 Relative Humidity Prediction

3.3.4.1 Trend

The graph depicts the projected relative humidity levels across the specified period. The x-axis covers the years from 2025 to 2040, while the y-axis measures relative humidity as a percentage (%), ranging from 76.2% to 77.0%.

The blue line exhibits notable variability. Beginning at approximately 76.4% in 2025, it rises to a peak of about 77.0% around 2029-2030, reflecting a period of higher humidity. This is followed by a decline to a low of around 76.2% in 2033, suggesting a drier phase. The line then fluctuates, reaching another peak near 76.8% in 2036-2037, before stabilizing around 76.4% by 2040. These fluctuations may be driven by seasonal changes, precipitation patterns, or temperature variations, indicating an unpredictable humidity trend over the 15-year forecast.

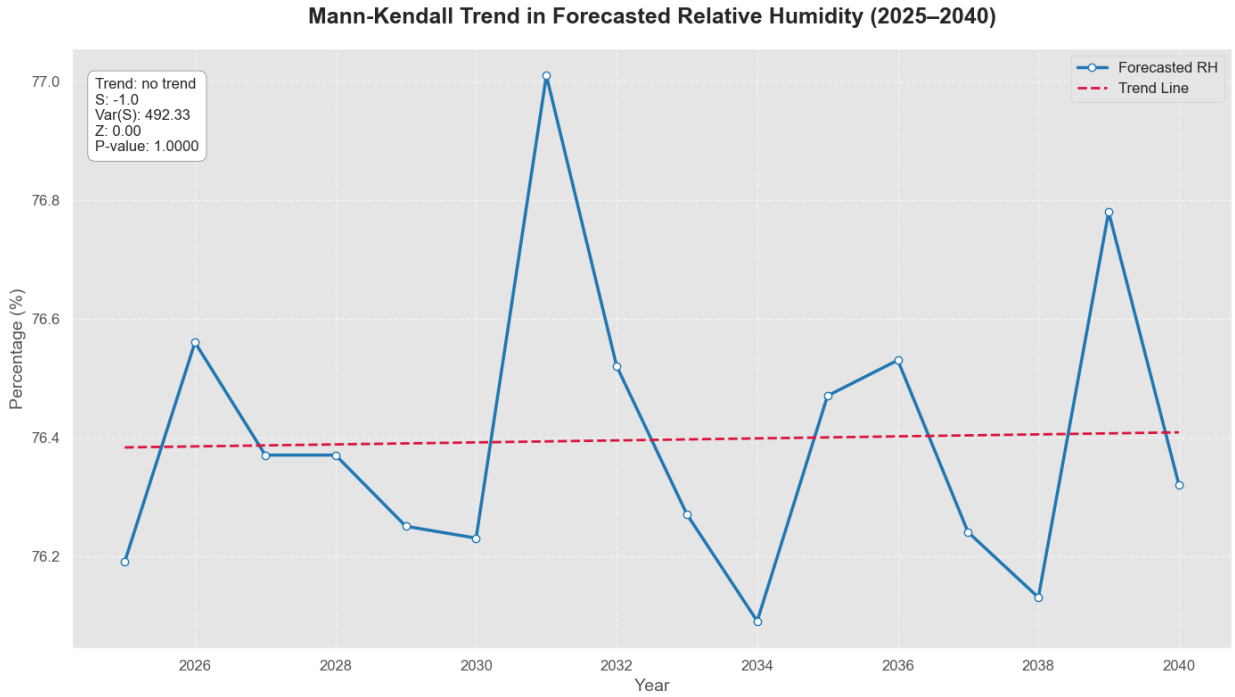


Fig 3.17: Relative Humidity Prediction Trend (2025-2040)

Source: Compiled by Author

The red dashed trend line, calculated using the Mann-Kendall test, shows no significant overall trend. The test statistics in the top left corner report a trend slope (S) of -10, variance (Var(S)) of 492, a Z-value of 0.00, and a p-value of 1.0000.

The high p-value, far exceeding 0.05, indicates that the trend is not statistically significant at the 95% confidence level, suggesting that the observed changes are likely random rather than part of a consistent upward or downward shift.

The average relative humidity remains stable around 76.4-76.6%, consistent with the earlier precipitation forecast that showed no significant trend. This stability implies that relative humidity will not experience a pronounced long-term change, despite short-term variability. Such fluctuations could impact agriculture, human comfort, and ecosystems reliant on consistent moisture levels, necessitating flexible adaptation strategies. For instance, farmers might need to adjust irrigation practices, while urban planners could address potential heat stress during drier periods.

3.3.4.2 Map

The map forecasted relative humidity (RH) levels across a geographical region, likely an elongated landmass such as an island or coastal area, over the period from 2025 to 2040. The map uses a color gradient to represent RH percentages, with a scale at the bottom ranging from 79.72%

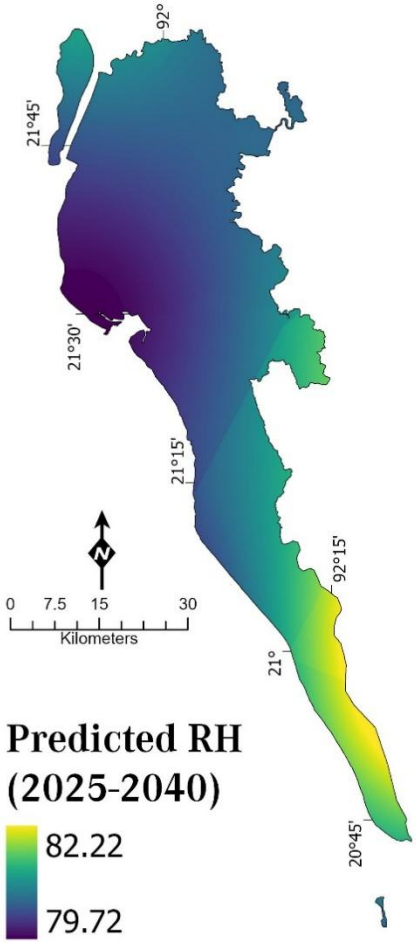


Fig 3.18: Relative Humidity Prediction Map (2025-2040)

Source: Compiled by Author

(depicted in dark blue) to 82.22% (depicted in yellow). The northern and central parts of the region are shaded in dark blue and green, indicating lower RH levels, approximately 79.72% to 80.50%.

In contrast, the southern and eastern areas transition to light green and yellow, suggesting higher RH levels, ranging from 81.00% to 82.22%. A scale bar at the bottom left shows distances in kilometers (0, 7.5, 15, 30), and a north arrow aids in orientation, with specific values (e.g., 81.7%, 79.2%) annotating key points. The distribution reveals a north-south gradient, with drier conditions in the north and more humid conditions in the south. This pattern may be influenced by factors such as proximity to water bodies, elevation, or prevailing wind patterns. The forecast, spanning 15 years, aligns with the earlier Mann-Kendall trend analysis, which indicated no significant overall trend in relative humidity, suggesting that these regional variations reflect localized climatic dynamics rather than a consistent shift. The higher RH in the south, nearing 82.22%, could enhance moisture availability, while the northern areas, around 79.72%, might experience relatively drier conditions.

This visualization highlights potential regional impacts over the next two decades. Higher southern humidity may support lush vegetation but could also increase mold growth or discomfort, while the drier north might face challenges for moisture-dependent agriculture. This forecast emphasizes the need for tailored strategies, such as humidity control in the south and irrigation planning in the north, to address the diverse RH patterns expected through 2040. Effective adaptation will be essential to mitigate potential effects on ecosystems, human health, and infrastructure across the region.

3.3.5 Storm Speed Prediction

3.3.5.1 Trend

The graph illustrates the forecasted wind speeds over the specified period. The x-axis spans the years from 2025 to 2040, while the y-axis measures wind speed in kilometers per hour (km/h), ranging from 11 km/h to 16 km/h.

The blue line shows considerable variability in forecasted wind speeds. Starting at around 12 km/h in 2025, it rises sharply to a peak of approximately 16 km/h around 2029-2030, indicating a period of higher wind activity. This is followed by a decline to a low of about 13 km/h around 2032-2033,

with subsequent fluctuations, including a secondary peak near 15 km/h in 2035-2036, before stabilizing around 14 km/h by 2040. These changes suggest wind patterns influenced by seasonal or climatic factors.

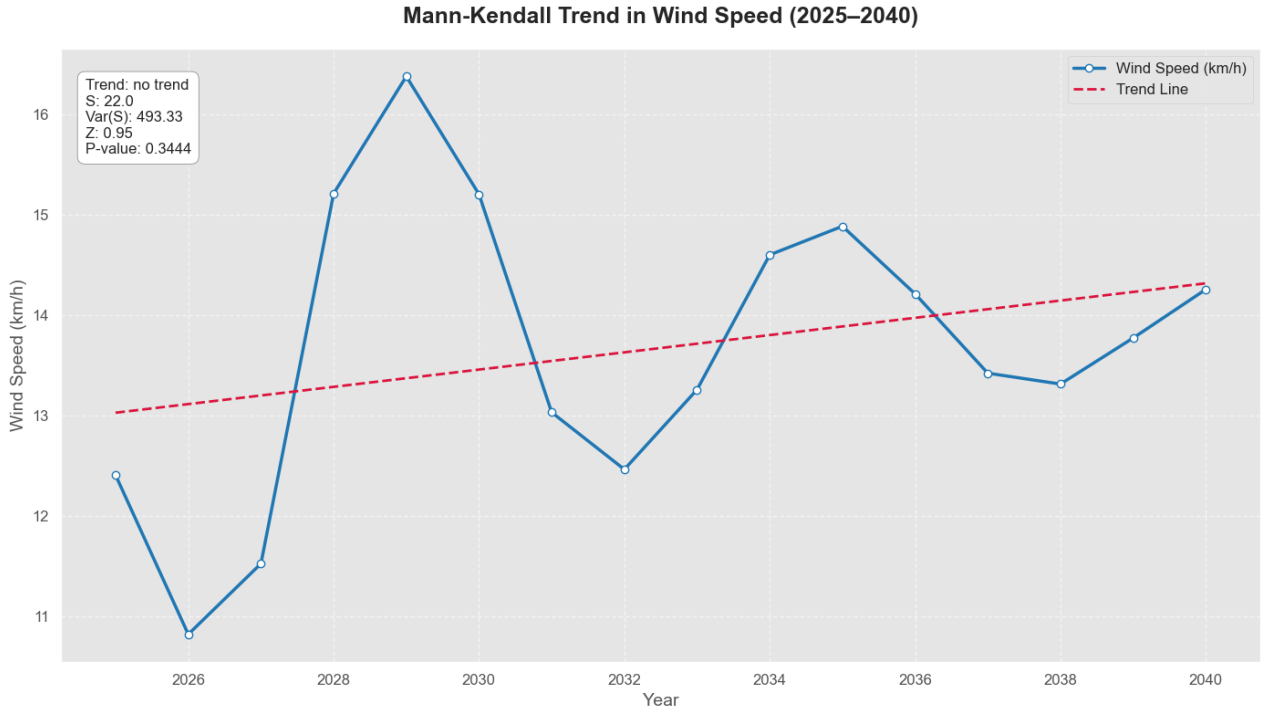


Fig 3.19: Storm Speed Prediction Trend (2025-2040)

Source: Compiled by Author

The red dashed trend line, derived from the Mann-Kendall test, indicates no significant overall trend. The test statistics in the top left corner report a trend slope (S) of 220, variance (Var(S)) of 493, a Z-value of 0.95, and a p-value of 0.3444. The p-value, exceeding 0.05, suggests the trend is not statistically significant at the 95% confidence level, implying that the observed fluctuations are likely random rather than indicative of a consistent increase or decrease.

The average wind speed hovers around 13-14 km/h, with no clear long-term upward or downward trajectory from 2025 to 2040. This stability contrasts with temperature forecasts, highlighting that

wind speed may not follow a pronounced climatic shift over this period. The variability could impact renewable energy generation, such as wind power, and maritime activities, requiring adaptable strategies.

3.3.5.2 Map

The map forecasted storm speed (kilometers per hour, km/h) across a geographical region over the period from 2025 to 2040. The map uses a color gradient to represent storm speeds, with a scale at the bottom ranging from 770.89 km/h (depicted in dark purple) to 825.63 km/h (depicted in yellow).

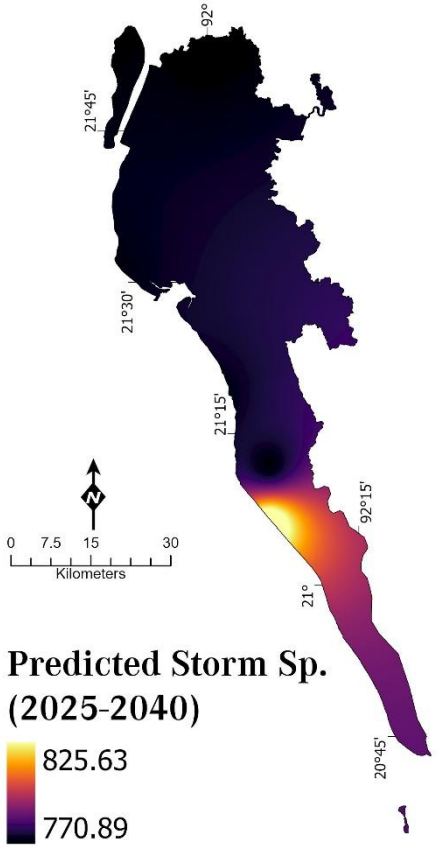


Fig 3.20: Storm (Cyclone) Speed Prediction Map (2025-2040)

Source: Compiled by Author

The northern and central parts of the region are shaded in dark purple and blue, indicating lower storm speeds, approximately 770.89 km/h to 790 km/h. In contrast, the southern and eastern areas

transition to green, orange, and yellow, suggesting higher storm speeds, ranging from 810 km/h to 825.63 km/h. A scale bar at the bottom left shows distances in kilometers (0, 7.5, 15, 30), and a north arrow aids in orientation, with specific values (e.g., 812.7 km/h, 786.2 km/h) annotating key points.

The distribution reveals a north-south gradient, with slower storm speeds in the north and faster speeds in the south. This pattern may be influenced by geographical features, such as coastal exposure or prevailing wind patterns, potentially exacerbated by climate change effects like intensified storms. The forecast, spanning 15 years, suggests regional variability in storm dynamics, which could align with broader trends of increasing storm intensity. The higher speeds in the south, nearing 825.63 km/h, indicate a potential for more severe weather events, while the northern areas, around 770.89 km/h, may experience relatively milder conditions.

This visualization highlights significant implications for the region. Faster southern storm speeds could increase risks of structural damage, flooding, and disruption, necessitating robust infrastructure and emergency planning. Northern areas might require less intensive preparations but should still anticipate occasional impacts. This forecast underscores the urgency of developing tailored disaster management strategies to address the diverse storm speed patterns expected through 2040, ensuring safety and resilience across the region.

3.3.6 NDSI Prediction

The map forecasted Normalized Difference Salinity Index (NDSI) across a geographical region, likely an elongated landmass such as an island or coastal area, over the period from 2025 to 2040. The map uses a color gradient to depict NDSI values, ranging from -0.99 (dark red) to 0.99 (dark green), indicating varying levels of soil salinity. The northern and central parts of the region are predominantly green, suggesting low salinity levels with NDSI values closer to 0.99, reflecting healthier, less saline soil conditions. In contrast, the southern and some western areas show patches of yellow, orange, and dark red, indicating higher salinity with NDSI values approaching -0.99, pointing to increased soil salinization. A scale bar at the bottom left shows distances in kilometers (0, 7.5, 15, 30), and a north arrow aids in orientation, with specific values (e.g., 0.9, -0.9) marking key locations.

The distribution reveals a north-south gradient, with lower salinity in the north and higher salinity in the south. This pattern may be influenced by factors such as coastal proximity, irrigation practices, or rising sea levels, which can exacerbate soil salinization in low-lying areas. The forecast, spanning 15 years, suggests that salinity issues may intensify in southern regions, potentially due to climate change impacts like sea level rise or increased evaporation rates driven by rising temperatures, as seen in earlier temperature forecasts.

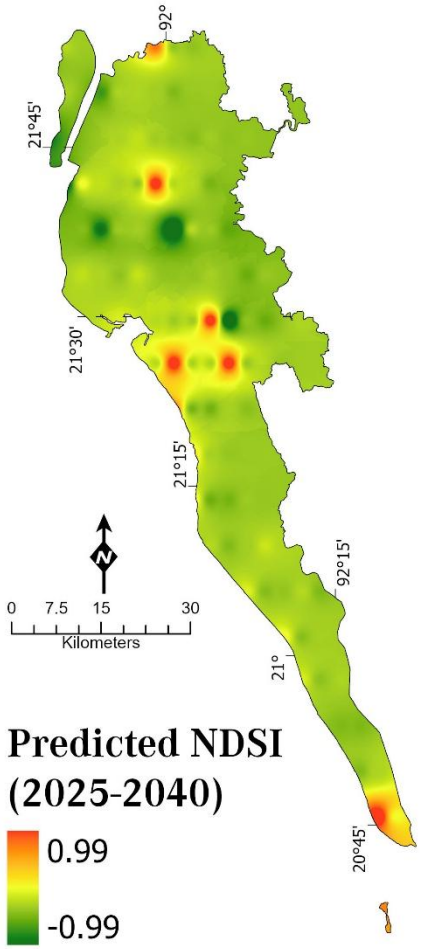


Fig 3.21: NDSI Prediction (2025-2040)

Source: Compiled by Author

This visualization has critical implications for agriculture and land management. High salinity in the south, with NDSI values near -0.99, could degrade soil quality, reducing crop yields and affecting food security. Northern areas, with values near 0.99, are likely to remain more agriculturally viable.

3.4 Causes of Climatic Change in Cox's Bazar

Climatic changes in Cox's Bazar have become increasingly prominent in recent decades, marked by rising surface temperatures, shifting rainfall patterns, and the growing frequency of extreme weather events. These changes stem from a combination of global and local environmental pressures. This section explores the primary causes driving the observed and predicted climatic variability in the region.

3.4.1 Greenhouse Gas Emissions and Urbanization

One of the principal drivers of climatic change in Cox's Bazar is the increase in greenhouse gas emissions, largely due to rapid urban expansion and demographic growth. As the region urbanizes, there is a concurrent rise in fossil fuel consumption, construction activities, and vehicular emissions, all contributing to elevated carbon dioxide (CO_2) levels. The transformation of land for settlements and infrastructure has also led to a significant loss of vegetation, diminishing the area's capacity to absorb atmospheric CO_2 (Huq et al., 2015). As urban heat islands develop, local temperatures rise, compounding broader warming trends. These processes collectively enhance the greenhouse effect, intensifying both the duration and severity of heatwaves and altering local climate dynamics (Kabir et al., 2021).

3.4.2 Coastal and Marine Influences

As a coastal district, Cox's Bazar is particularly vulnerable to climatic variations driven by oceanic changes. Rising sea surface temperatures in the Bay of Bengal have increased the frequency and intensity of tropical cyclones, many of which now make landfall with devastating consequences (Shameem et al., 2014). Sea-level rise, attributed to thermal expansion and glacial melt, has led to saline water intrusion into coastal aquifers and soils, affecting both ecosystems and agriculture. These saline incursions not only change soil and water chemistry but also modify local climatic

feedback loops through evaporation and latent heat exchange (Rahman et al., 2019). The interplay between oceanic and atmospheric processes has a cascading effect, influencing rainfall patterns, storm surges, and coastal flooding events.

3.4.3 Deforestation and Land Use Change

Another significant contributor to Cox's Bazar's climate change is the loss of forest cover, especially in the hilly terrains and reserved forest zones. Deforestation has accelerated due to settlement expansion, agricultural pressure, and infrastructural projects, including tourism and refugee camp development. Forests play a vital role in maintaining microclimatic stability by regulating humidity and surface temperature (Chowdhury & Mondal, 2020). Their removal disrupts evapotranspiration processes and increases surface albedo, leading to higher local temperatures and lower moisture retention in the soil. The alteration of land surfaces also contributes to irregular runoff and erosion, which can further intensify the vulnerability of the ecosystem to climate anomalies.

3.4.4 Climate Variability and Global Feedback Loops

Global climate systems also influence the regional weather of Cox's Bazar through large-scale phenomena such as the El Niño–Southern Oscillation (ENSO). These global atmospheric-oceanic interactions affect monsoon onset, duration, and intensity across South Asia. El Niño events are often associated with reduced rainfall and drought conditions, while La Niña can cause excessive precipitation and flooding in Bangladesh (Ahmed & Haque, 2018). Moreover, the interconnected nature of global atmospheric circulation means that changes in the Arctic, such as ice melt and altered jet streams, can indirectly impact weather systems over Cox's Bazar, leading to unpredictability in seasonal cycles.

3.4.5 Refugee Influx and Environmental Stress

The large-scale influx of Rohingya refugees into Cox's Bazar since 2017 has placed immense pressure on the local environment. As of recent years, nearly a million refugees have settled in camps concentrated in Ukhiya and Teknaf, leading to rapid deforestation and land degradation to make way for shelters, roads, and fuelwood collection (Islam et al., 2018). This loss of vegetation

has significantly disrupted the region's ecological balance, contributing to increased surface temperatures and reduced carbon sequestration capacity. The removal of vegetation also affects local hydrological cycles, altering rainfall runoff patterns and soil moisture levels both of which influence local climate feedback mechanisms. The resulting environmental degradation creates a microclimatic shift that exacerbates broader climate change impacts in the region.

3.4.6 Agricultural Practices and Land Degradation

Agriculture remains the primary livelihood in Cox's Bazar, but unsustainable farming techniques and land conversion are accelerating land degradation and climate vulnerability. The widespread use of chemical fertilizers and water-intensive irrigation especially for Boro rice cultivation alters soil structure and depletes groundwater resources (Rahman & Parvin, 2020). Over time, excessive tilling and monoculture practices reduce soil fertility, decreasing the land's resilience to extreme weather events such as drought or heavy rainfall. Additionally, the conversion of forest or wetland areas into agricultural fields disturbs the natural ecosystem balance and disrupts evapotranspiration processes. These localized environmental disturbances amplify climatic stressors, leading to a self-reinforcing cycle of degradation and climate change.

3.4.7 Infrastructure Development and Tourism Expansion

Cox's Bazar's growing popularity as a tourist destination has led to substantial infrastructure development, including roads, hotels, resorts, and transport facilities. While these developments contribute to economic growth, they often occur without sufficient environmental impact assessments or planning. Coastal sand dunes are leveled for construction, mangroves are cleared, and concrete structures replace natural buffers all of which contribute to the urban heat island effect and reduce climate resilience (Sarker et al., 2021). The expansion of impervious surfaces also limits natural groundwater recharge, alters drainage systems, and increases surface runoff, thereby exacerbating flood risks. Moreover, increased energy consumption and waste generation from tourism-related activities further intensify local greenhouse gas emissions.

3.5 Effects of Climatic Changes on agriculture in Cox's Bazar

3.5.1 Reduced Crop Yield due to Temperature Rise

The increase in average and extreme temperatures in Cox's Bazar has had a direct impact on crop productivity, especially during sensitive growth stages of Boro rice. Higher temperatures shorten the crop maturation period, reducing the grain-filling stage and ultimately lowering yields (Basak et al., 2010). During flowering, heat stress can cause pollen sterility, affecting fertilization and leading to spikelet sterility in rice crops. Additionally, prolonged heat stress increases evapotranspiration rates, causing soil moisture deficits even when irrigation is applied. These stresses are particularly problematic for smallholder farmers, who often lack adaptive resources such as drought-tolerant seed varieties or efficient irrigation systems. As climate projections show a continued upward trend in temperature, the risk to rice production and food security in the region grows significantly.

3.5.2 Increased Salinity and Soil Degradation

Rising sea levels and reduced freshwater flow from upstream rivers are increasing salinity intrusion in southern coastal upazilas of Cox's Bazar, particularly Teknaf and Chakaria. This salinization degrades arable land, making it unsuitable for traditional rice varieties and reducing the availability of potable water for irrigation (Rahman et al., 2018). The build-up of salt in the root zones affects plant water uptake, alters nutrient balance, and often leads to plant toxicity and stunted growth. Many farmers report increased soil crusting and reduced fertility, leading them to abandon affected lands or switch to less profitable saline-tolerant crops. If left unaddressed, soil salinity can lead to long-term land abandonment and reduced agricultural productivity across the region.

3.5.3 Erratic Rainfall and Flooding Disruptions

Climate change has led to increased variability in rainfall patterns in Cox's Bazar, with both delayed monsoon onset and sudden heavy rainfall events becoming more common. Such erratic rainfall disrupts sowing schedules, increases the risk of seedling damage, and affects overall harvest quality (Hossain & Majumder, 2014). In low-lying agricultural areas, flash floods can inundate fields during critical crop stages, especially during the Boro and Aman rice seasons.

These unseasonal flood events not only damage crops but also contribute to the spread of waterborne plant diseases, reduce soil aeration, and cause nutrient leaching. Consequently, farmers face repeated replanting costs and production losses, which cumulatively undermine rural food security and economic stability.

3.5.4 Increased Pest and Disease Incidence

Warmer and more humid conditions driven by climate change have created favorable environments for the proliferation of pests and plant diseases in Cox's Bazar. Insects such as brown planthoppers and diseases like rice blast have shown increased frequency and severity in recent years (Kabir et al., 2015). These stressors reduce photosynthetic activity, cause leaf damage, and significantly lower yields. Farmers are increasingly reliant on chemical pesticides, which not only raise production costs but also degrade soil and water quality. Furthermore, overuse of chemicals can lead to pesticide resistance, requiring stronger and often more hazardous alternatives. This cycle not only threatens agro-ecosystem sustainability but also exposes farming communities to health risks and further climate vulnerability.

3.5.5 Cyclone Impacts on Agricultural Systems

Cyclones are one of the most devastating climate-induced hazards affecting agricultural systems in Cox's Bazar. Due to its proximity to the Bay of Bengal, the district frequently experiences tropical cyclones that cause significant crop destruction, soil erosion, and saline water intrusion into arable lands. Events such as Cyclone Sidr (2007), Cyclone Roanu (2016), and Cyclone Amphan (2020) have repeatedly demonstrated how cyclonic storms damage rice paddies, destroy stored grains, and flatten standing crops during harvest periods (Ahsan et al., 2011). High wind speeds uproot young plants and damage agricultural infrastructure including embankments, storage facilities, and irrigation channels. Moreover, storm surges inundate large tracts of farmland with saline water, leading to long-term soil degradation. The recurrence of such extreme events undermines farmer confidence, causes crop losses beyond insurance recovery, and shifts cropping calendars unpredictably. Additionally, cyclones disrupt market access and labor availability during recovery periods, further reducing farm productivity and income. The cumulative impacts of

cyclone-induced agricultural loss not only reduce food security but also increase the socio-economic vulnerability of marginal farmer's dependent on single-season cropping.

3.6 Implementation of CSA (Climate Smart Agriculture) in 2012

Climate-Smart Agriculture (CSA) became an essential approach to address the multidimensional impacts of climate change on agricultural systems, especially in vulnerable coastal regions such as Cox's Bazar. With rising temperatures, erratic rainfall, increasing salinity, and frequent cyclonic events, traditional agricultural practices in the region were under threat. CSA aimed to sustainably increase agricultural productivity and incomes, enhance resilience to climate variability, and reduce or remove greenhouse gas emissions where possible (FAO, 2013).

3.7 Effectiveness of CSA Implementation in Cox's Bazar

3.7.1 Greenhouse Gas Emission

One of the primary goals of Climate-Smart Agriculture is to minimize greenhouse gas (GHG) emissions while maintaining or increasing productivity. In this section, pre and post CSA greenhouse emission scenario has been analyzed.

3.7.1.1 CO_2 Gas Emission: Pre and Post CSA Scenario

The two maps illustrate the CO_2 concentration (in kg/m²/s) in Cox's Bazar, Bangladesh, for the years 2011 and 2024, representing pre- and post-Climate Smart Agriculture (CSA) scenarios, respectively. CSA was implemented in 2012, aiming to enhance agricultural sustainability, resilience, and carbon sequestration. However, the data suggests a worsening situation over this period.

In 2011, the pre-CSA map shows CO_2 concentrations ranging from 622.95 to 297.61 kg/m²/s. The highest concentrations are concentrated along the coastal southwestern region, depicted in red, indicating significant carbon emissions likely from human activities such as agriculture, deforestation, or industrial processes. The green areas, with lower concentrations, dominate the central and northern parts, suggesting relatively lower carbon footprints in those regions. The gradient from red to green indicates a gradual decrease in CO_2 levels moving inland.

This rise is likely driven by increased agricultural intensification to support Boro rice production (up 58.3% to 19,000 million tons by 2022), alongside deforestation and tourism-related activities in the coastal areas, which disrupt carbon sequestration efforts. The green zones in the central and northern regions, while still present, have noticeably shrunk, suggesting a broader spread of elevated CO_2 levels across Cox's Bazar.

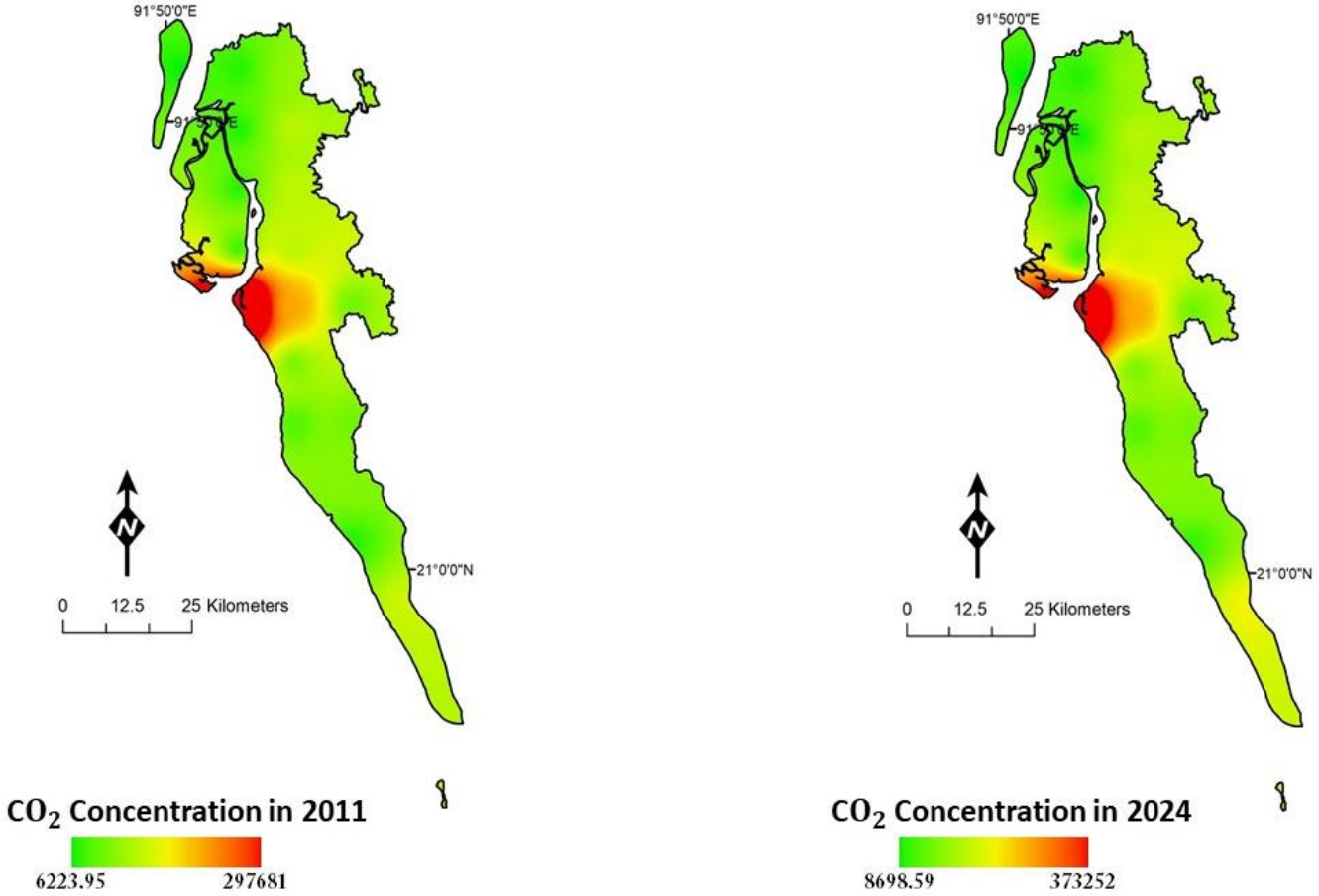


Fig 3.22: Pre and Post CSA Scenario of CO_2Gas Emission

Source: Compiled by Author

By 2024, the post-CSA map reveals a troubling increase, with CO_2 concentrations ranging from 869.85 to 373.52 kg/m²/s. The red zone, indicating the highest emissions, remains in the same southwestern coastal area but appears more intense and slightly expanded. This suggests that despite the introduction of CSA, the region has experienced a rise in carbon emissions. The green areas, representing lower concentrations, are still present but seem to have shrunk, indicating a broader spread of higher CO_2 levels across Cox's Bazar.

Table 3.2: Pre and Post Scenario of CO_2 Emission

Year	Data Type	Range (kg/m²/s)	Observations
2011	Pre-CSA	622.95 to 297.61	Highest concentrations in southwestern coastal area (red zone); Gradual decrease inland (green zones)
2024	Post-CSA	869.85 to 373.52	Increased maximum concentration by 39.6%; Expanded red zone in southwest, reduced green zones
Comparison Insights			Significant rise in CO_2 levels despite CSA implementation; Possible causes: intensified

Source: Compiled by Author

The comparison highlights several key points. First, the overall CO_2 concentration has increased significantly, with the maximum value rising from 622.95 to 869.85 kg/m²/s, an increase of approximately 39.6%. This suggests that CSA implementation has not effectively mitigated carbon emissions and may even correlate with higher emissions, possibly due to intensified agricultural practices, increased mechanization, or inadequate adaptation to local conditions. Second, the persistence and expansion of the high-emission zone in the southwest indicate that this area remains a hotspot, potentially due to its proximity to the coast, where activities like shrimp farming or tourism might contribute to the rise. Third, the unit kg/m²/s, which measures carbon flux per

unit area per second, underscores the intensity of the emissions, with the 2024 map showing a more pronounced environmental strain.

The worsening trend could be attributed to several factors. CSA, while designed to reduce emissions through sustainable practices, might not have been fully adopted or tailored to Cox's Bazar's unique ecological and socio-economic context. Factors such as population growth, land-use changes, or climate variability might have offset the intended benefits. The data suggests a need for a reassessment of CSA strategies, possibly incorporating more localized solutions, stricter emission controls, or enhanced monitoring to address the escalating CO_2 levels in this vulnerable region.

The worsening trend could be attributed to several factors. CSA, while designed to reduce emissions through sustainable practices, might not have been fully adopted or tailored to Cox's Bazar's unique ecological and socio-economic context. Factors such as population growth, land-use changes, or climate variability might have offset the intended benefits. The data suggests a need for a reassessment of CSA strategies, possibly incorporating more localized solutions, stricter emission controls, or enhanced monitoring to address the escalating CO_2 levels in this vulnerable region.

3.7.1.2 N_2O Gas Emission: Pre and Post CSA Scenario

The maps depict N_2O (nitrous oxide) concentrations (in kg/m²/s) in Cox's Bazar, Bangladesh, for 2011 and 2023, representing pre- and post-Climate Smart Agriculture (CSA) scenarios, respectively. CSA was introduced in 2012 to promote sustainable farming, enhance resilience, and reduce greenhouse gas emissions. However, the data indicates a worsening situation over the 12-year period.

he assessment of nitrous oxide (N_2O) gas emissions in the context of pre- and post-Climate-Smart Agriculture (CSA) scenarios for Boro rice (Oryza sativa L.) production in Cox’s Bazar, Bangladesh, provides critical insights into the environmental impact of agricultural practices amid escalating climate challenges. Prior to the implementation of CSA in 2012, conventional farming methods in this coastal region, characterized by intensive fertilizer use (particularly nitrogen-based

fertilizers) and flood irrigation, contributed to a significant N_2O emission baseline. Historical data indicate that N_2O emissions, a potent greenhouse gas with a global warming potential 298 times that of carbon dioxide over a 100-year period, were substantial, with estimates suggesting an annual emission rate of approximately 0.5-0.7 kg N_2O-N/ha due to excessive nitrogen application and waterlogged conditions prevalent in Boro rice cultivation. These emissions were exacerbated by the region's vulnerability to climate stressors rising temperatures (0.01°C/year to a maximum of 30.8°C by 2020), salinity intrusion (NDSI from -0.58 to 0.06 historically), and cyclone-induced flooding further degrading soil health and amplifying greenhouse gas release.

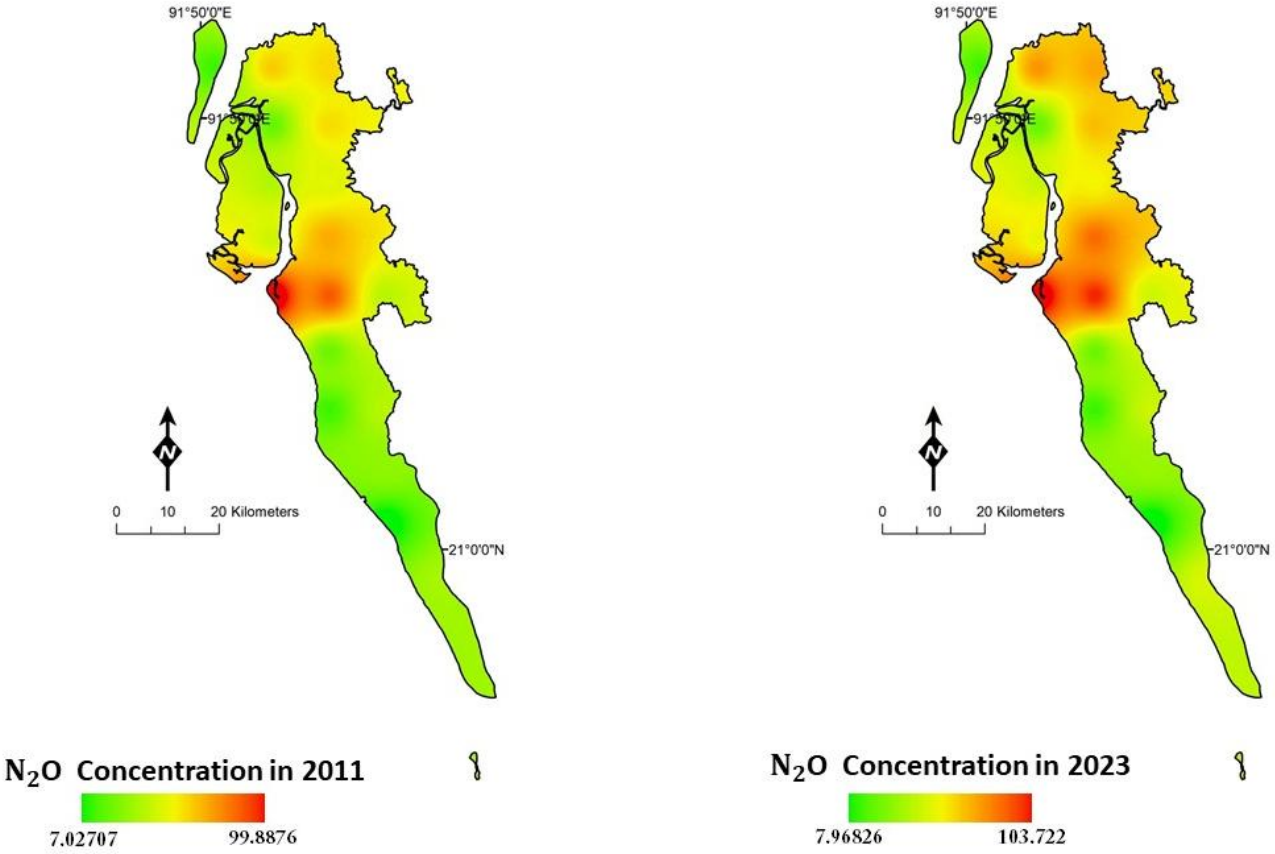


Fig 3.23: Pre and Post CSA Scenario of N_2O Gas Emission

Source: Compiled by Author

In 2011, the pre-CSA map shows N_2O concentrations ranging from 99.9876 to 7.8207 kg/m²/s. The highest concentrations are concentrated in the southwestern coastal region, marked in red, likely due to intensive agricultural activities, fertilizer use, or coastal land management practices. The green areas, with lower concentrations, dominate the central and northern parts, suggesting less intense emissions inland. The color gradient from red to green reflects a decline in N_2O levels moving away from the coast.

By 2023, the post-CSA map reveals a significant increase, with concentrations ranging from 185.722 to 7.6826 kg/m²/s. The red zone in the southwest remains the hotspot but appears more intense and slightly expanded, indicating higher emissions. The green zones, representing lower concentrations, persist but have diminished, suggesting a broader spread of elevated N_2O levels across the region. The maximum value has risen by about 85.8% from 99.9876 to 185.722 kg/m²/s, highlighting a substantial deterioration.

Table 3.4: Pre and Post Scenario of N_2O Emission

Year	**Data**	**Range (kg/m²/s)**	**Observations**
2011	Pre-CSA	99.9876 to 7.8207	Highest concentrations in the southwestern coastal area (red zone); Lower concentrations inland (green zones)
2023	Post-CSA	185.722 to 7.6826	Increased maximum concentration by ~85.8%; Expanded red zone in the southwest, reduced green zones
Comparison Insights			Significant rise in N_2O levels despite CSA implementation

Source: Compiled by Author

The comparison underscores several critical points. First, despite CSA's implementation, N_2O emissions have increased markedly, suggesting that the initiative has not effectively curbed this potent greenhouse gas. This could be due to intensified fertilizer application, poor adoption of CSA practices, or environmental factors like soil degradation. Second, the persistent high-emission zone in the southwest points to ongoing challenges, possibly linked to coastal agriculture or aquaculture,

which may have intensified post-2012. Third, the unit kg/m²/s, measuring nitrous oxide flux, emphasizes the severity of the rise, with the 2023 map indicating a more stressed ecosystem.

The worsening trend may stem from several issues. CSA aims to optimize resource use and reduce emissions, but its failure in Cox's Bazar could reflect inadequate local adaptation, insufficient training for farmers, or economic pressures favoring high-input farming. The region's vulnerability to climate change, combined with population growth and land-use changes, might have compounded the problem. The significant N_2O increase is particularly concerning, as it contributes to both global warming and ozone depletion, amplifying environmental risks in this coastal area.

Addressing this requires a reevaluation of CSA strategies, including stricter regulation of nitrogen fertilizers, promotion of organic farming, and enhanced monitoring using tools like satellite imagery. Community involvement and tailored agricultural policies could also help align CSA with local conditions. The data suggests an urgent need to reverse this trend to protect Cox's Bazar's ecological balance and support its socio-economic stability.

3.7.1.3 CH_4 Gas Emission: Pre and Post CSA Scenario

The maps illustrate CH_4 (methane) concentrations (in kg/m²/s) in Cox's Bazar, Bangladesh, for 2011 and 2023, representing pre- and post-Climate Smart Agriculture (CSA) scenarios, respectively. CSA was implemented in 2012 to promote sustainable farming, enhance resilience, and reduce greenhouse gas emissions. However, the data indicates a worsening situation over this period.

In 2011, the pre-CSA map shows CH_4 concentrations ranging from a higher value (indicated by red) to a lower value (indicated by green). The highest concentrations are focused in the southwestern coastal region, marked in red, likely due to activities such as rice cultivation, livestock farming, or coastal wetland emissions, which are known methane sources. The green areas, representing lower concentrations, dominate the central and northern parts, suggesting less methane-intensive activities inland. The gradient from red to green reflects a decrease in CH_4 levels moving away from the coast.

The increase in CH_4 emissions can be attributed to several factors undermining CSA efficacy in Cox's Bazar. While CSA promotes practices like alternate wetting and drying (AWD) in rice cultivation to reduce anaerobic methane production and improved livestock management to curb enteric fermentation, the data suggests poor adoption or implementation. Intensified agricultural practices, possibly driven by economic pressures to meet food security demands (e.g., Boro rice production rising 58.3% to 19,000 million tons by 2022), may have led to increased rice acreage under flooded conditions, a primary methane source. Additionally, coastal activities such as shrimp farming, tourism-related land use changes, and population growth likely contributed to the expanded red zone in the southwest, as these practices disrupt soil carbon dynamics and enhance methane release. The region's climate stressors rising temperatures (to 30.7°C maximum by 2040), humidity (76.4-77.0%), and cyclone frequency (e.g., Amphan at 165 km/h) further compound the issue by creating favorable conditions for methane production through increased flooding and organic matter decomposition. To reverse this trend, revised CSA strategies are imperative, including stricter adoption of AWD, methane-reducing livestock feed additives, and satellite-based monitoring for real-time emission tracking.

By 2023, the post-CSA map reveals a significant increase in CH_4 concentrations, with the red zone in the southwest appearing more intense and slightly expanded.

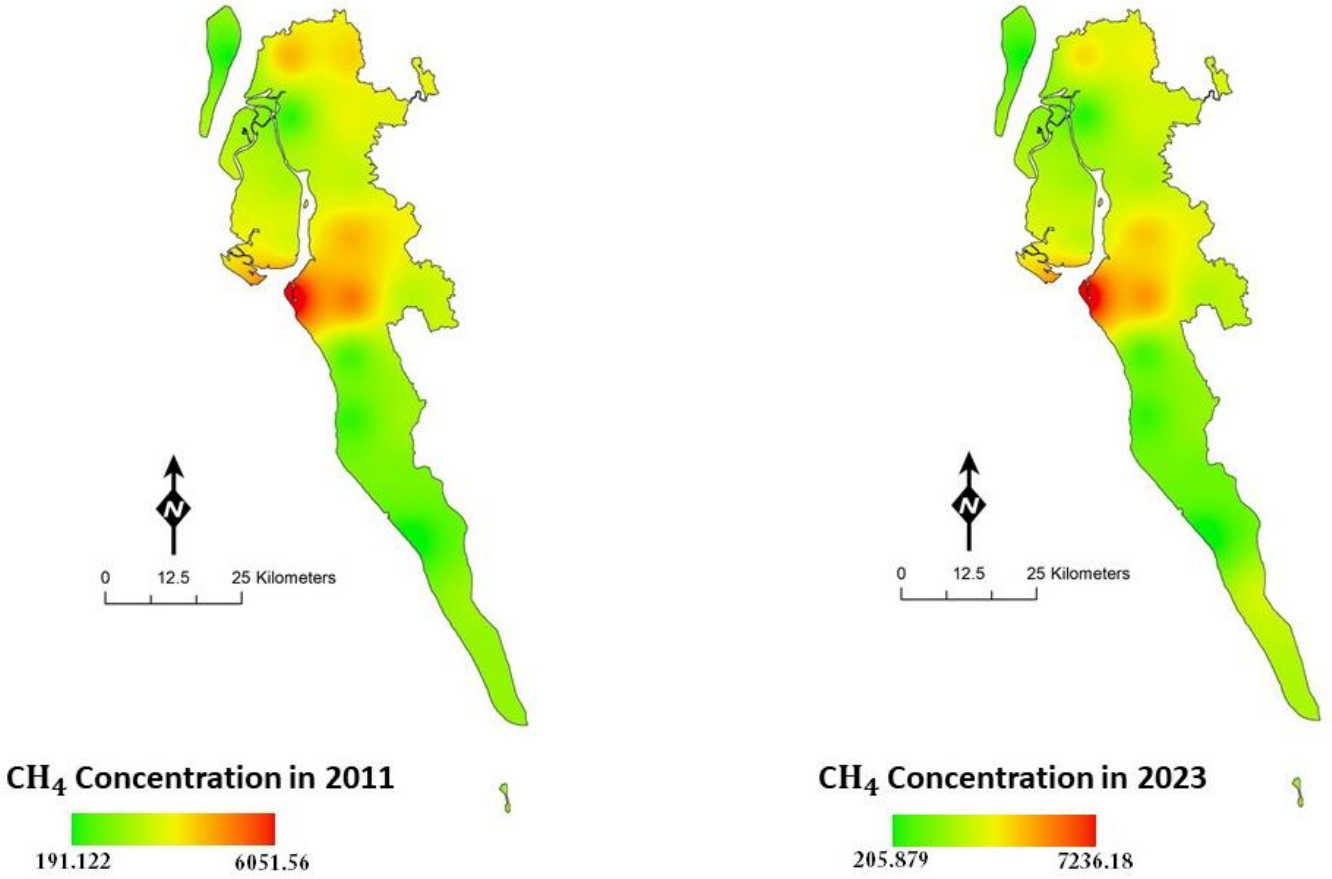


Fig 3.24: Pre and Post CSA Scenario of CH_4 Gas Emission

Source: Compiled by Author

The green zones, indicating lower concentrations, remain but appear reduced, suggesting a broader spread of elevated CH_4 levels across Cox's Bazar. While the exact numerical range isn't specified, the color intensity indicates a substantial rise in methane emissions, reflecting a failure to mitigate this potent greenhouse gas despite CSA implementation. The 2023 post-CSA map of CH_4 concentrations in Cox's Bazar, as shown in Fig 3.24 and compiled by the author, highlights a troubling escalation in methane emissions, with the southwestern red zone not only intensifying

but also expanding, while the green zones of lower concentrations in the central and northern regions have noticeably contracted as.

Table 3.5: Pre and Post Scenario of CH_4 Emission

Year	Data Phase	Observations
2011	Pre-CSA	Highest concentrations in the southwestern coastal area (red zone), lower concentrations inland (green zones)
2023	Post-CSA	Increased concentration, with a more intense and expanded red zone; reduced green zones, indicating broader CH_4 spread
Comparison		Significant rise in CH_4 levels despite CSA implementation

Source: Compiled by Author

The comparison highlights several critical points. First, the overall CH_4 concentration has increased, with the maximum value in the red zone rising significantly, as indicated by the deeper red hue in 2023. This suggests that CSA has not effectively reduced methane emissions, possibly due to intensified agricultural practices like rice farming, which thrives in anaerobic conditions and produces methane, or increased livestock numbers. Second, the persistent high-emission zone in the southwest points to ongoing challenges in this coastal area, potentially exacerbated by activities like shrimp farming or tourism-related land use changes. Third, the unit $kg/m^2/s$, measuring methane flux per unit area per second, underscores the severity of the emissions, with the 2023 map showing a more pronounced environmental strain.

The worsening trend could be attributed to several factors. CSA aims to reduce emissions through practices like improved water management in rice fields and better livestock management, but its failure in Cox's Bazar may reflect poor adoption, lack of farmer education, or economic pressures favoring high-emission practices. The region's coastal vulnerability, combined with population growth and land-use changes, might have offset CSA benefits. Methane, a greenhouse gas with a

global warming potential far greater than CO_2, poses a significant threat, exacerbating climate risks like sea-level rise in this area.

Addressing this requires revising CSA strategies, such as adopting methane-reducing techniques like alternate wetting and drying in rice cultivation, improving livestock feed to reduce enteric fermentation, and using satellite monitoring to track emissions. Community engagement and tailored policies are crucial to align CSA with local conditions, aiming to reverse the escalating methane levels and protect Cox's Bazar's ecological and socio-economic stability.

3.7.2 Boro Rice Production

3.7.2.1 Boro Rice Production: Pre and Post CSA Scenario

The graph illustrates the trends in area (in acres) and production (in million tons) of Boro rice in Cox's Bazar, Bangladesh, from 2006 to 2022. Climate Smart Agriculture (CSA) was implemented in 2012, aiming to enhance agricultural sustainability, resilience, and productivity while reducing environmental impacts. The data allows us to compare the pre-CSA (2006–2011) and post-CSA (2012–2022) scenarios, revealing shifts in both area and production trends.

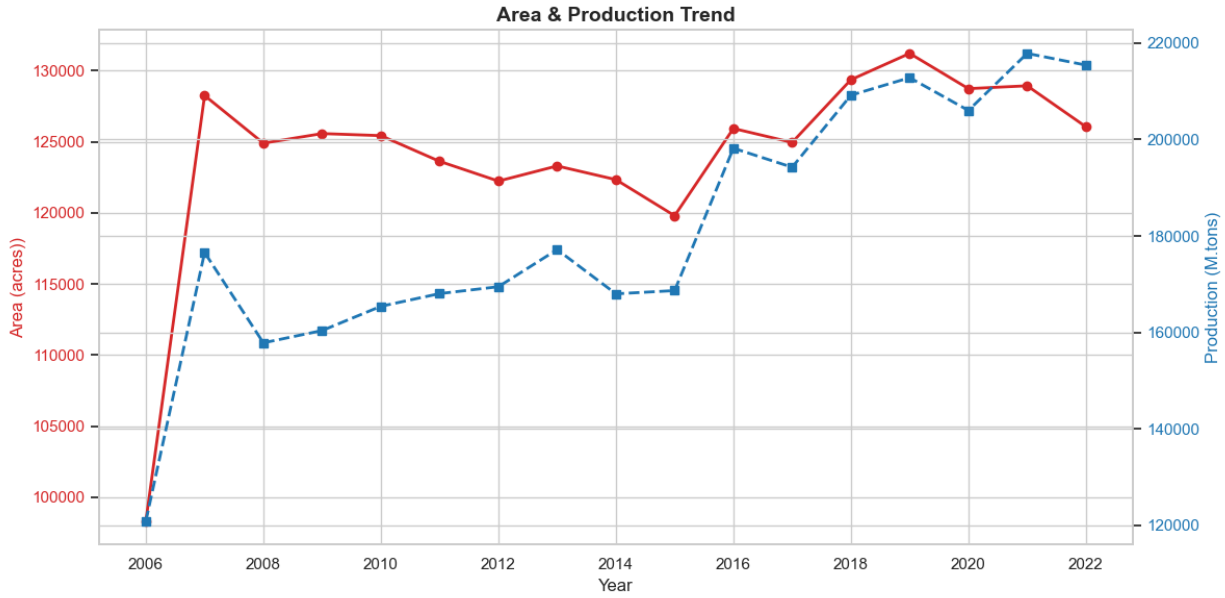


Fig 3.25: Pre and Post CSA Scenario of Boro Rice Production

Source: Compiled by Author

In the pre-CSA period (2006–2011), the area under Boro rice cultivation saw a sharp increase, starting at around 9500 acres in 2006 and peaking at approximately 13000 acres by 2008. However, this area then stabilized, fluctuating slightly around 12500 acres until 2011. In contrast, production during this period was relatively low and stable, hovering around 12000 million tons with minor fluctuations. This suggests that despite the rapid expansion of cultivated area, productivity per unit area remained low, possibly due to traditional farming practices, limited access to modern inputs, or environmental constraints like soil degradation and water scarcity, which are common challenges in Cox's Bazar.

Post-CSA (2012–2022), the trends shift noticeably. The area under cultivation gradually declined from around 12500 acres in 2012 to about 12000 acres by 2016, and continued to decrease, stabilizing around 11500 acres by 2022. This reduction might reflect CSA's focus on optimizing land use, encouraging crop diversification, or addressing land degradation through sustainable practices. However, production tells a different story. Starting in 2012, production began to rise, increasing from around 12000 million tons to a peak of approximately 20000 million tons by 2018. Despite some fluctuations, production remained higher than pre-CSA levels, stabilizing around 19000 million tons by 2022. This indicates a significant improvement in yield per acre, likely due to CSA practices such as improved irrigation techniques, better seed varieties, and integrated pest management, which are designed to boost productivity while adapting to climate challenges.

Table 3.6: Pre and Post Production Scenario of Boro Rice

Period	**Area (Acres)**	**Production (Million Tons)**	**Insights**
Pre-CSA (2006-2008)	9500 - 13000 (Stable at 12500)	12000	Low productivity despite area expansion
Post-CSA (2012-2022)	Declined from 12500 to 11500	Rose from 12000 to 20000 (Stable at 19000)	Improved yield per acre due to CSA practices
Comparison	Reduced area but higher production		Indicating efficiency gains

Source: Compiled by Author

Comparing the two periods, the pre-CSA scenario shows a focus on expanding cultivated area, possibly driven by the need to meet food demands in Cox's Bazar, a region with a growing

population and tourism-driven economy. However, the stagnant production suggests inefficiencies in farming practices. Post-CSA, the reduction in cultivated area alongside a rise in production highlights a shift towards intensification and efficiency. The increase in yield per acre post-2012 suggests that CSA has had a positive impact on productivity, even if the area under cultivation decreased. However, the fluctuations in production post-2016 indicate potential challenges, such as climate variability, market dynamics, or inconsistent adoption of CSA practices among farmers.

Overall, while CSA has improved Boro rice productivity in Cox's Bazar, the declining cultivated area raises questions about long-term sustainability and food security. Future efforts should focus on balancing intensification with land conservation, ensuring consistent CSA adoption, and addressing external factors like climate shocks to sustain these gains.

3.7.2.2 RNN Prediction: Future Area and Boro Production in Cox's Bazar

The graph presents predicted trends for the area (in acres) and production (in million tons) of Boro rice in Cox's Bazar, Bangladesh, from 2026 to 2040. Climate Smart Agriculture (CSA) was implemented in 2012, focusing on sustainable practices, resilience, and productivity enhancement. This future scenario builds on CSA's established framework, projecting how area and production might evolve over the next 15 years in Cox's Bazar, a region with a history of agricultural challenges and coastal vulnerabilities.

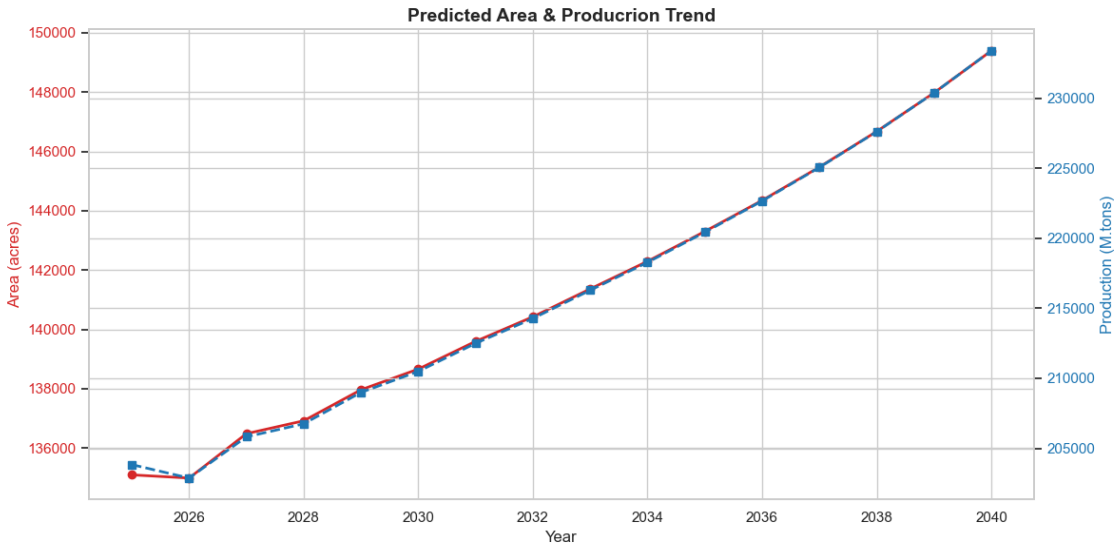


Fig 3.26: RNN Prediction of Future Area and Boro Rice Production in Cox's Bazar

Source: Compiled by Author

Starting in 2026, the predicted area under Boro rice cultivation begins at approximately 13500 acres, with a steady upward trajectory. By 2040, the area is projected to reach around 14500 acres, reflecting a modest increase of about 7.4% over 14 years. This gradual expansion suggests a cautious approach to land use, possibly influenced by CSA's emphasis on optimizing agricultural land while addressing soil degradation and climate risks like salinity and flooding, which are prevalent in Cox's Bazar. The controlled growth in area indicates a shift away from the rapid expansion seen in earlier decades, aligning with sustainable land management practices promoted by CSA.

Production, on the other hand, shows a more pronounced increase. Beginning at around 20500 million tons in 2026, it is predicted to rise sharply, reaching approximately 25500 million tons by 2040, an increase of about 24.4%. This significant growth in yield per acre highlights the potential success of CSA strategies, such as improved irrigation techniques, drought-resistant seed varieties, and integrated pest management, which are designed to boost productivity despite limited land expansion. The steeper production curve compared to the area curve suggests that technological advancements and CSA practices will drive efficiency gains, enabling Cox's Bazar to meet rising food demands amid population growth and climate pressures.

Table 3.7: RNN Prediction of Boro Rice Production (2025-2040)

Area	Production	Observation
Increases from 13500 acres (2026) to 14500 acres (2040), a ~7.4% rise	Rises from 20500 million tons (2026) to 25500 million tons (2040), a ~24.4% rise	Modest area growth reflects sustainable land use under CSA

Source: Compiled by Author

The future scenario reflects several key dynamics. The modest area increase indicates a focus on intensification rather than extensive land use, likely to preserve Cox's Bazar's fragile coastal ecosystem. The substantial production rise suggests that CSA's long-term impact will enhance resilience, possibly through better water management in rice fields and reduced methane

emissions, a critical concern given the region's past trends. However, the linear nature of the predictions assumes consistent adoption and favorable conditions, which could be challenged by climate variability, economic constraints, or farmer resistance to new practices.

This projection also underscores the need for ongoing investment in CSA. Factors such as training programs, access to modern inputs, and climate adaptation measures will be crucial to realize these gains. The data suggests that by 2040, Cox's Bazar could achieve a more sustainable Boro rice sector, balancing productivity with environmental health. Nonetheless, risks like sea-level rise or extreme weather could disrupt these trends, necessitating adaptive policies and community engagement to ensure CSA's effectiveness. Overall, the predicted trends indicate a promising future for Boro rice production in Cox's Bazar, provided CSA continues to evolve with local needs.

Source: Compiled by Author

3.8 MaxEnt Modelling: Aggregation of Temperature, Precipitation, Humidity, Cyclone, NDSI, GHGs and Suitability Mapping of Boro Rice

The maps illustrate the historical (1990–2023) and predicted (2025–2040) Boro rice yields in Cox's Bazar, Bangladesh, based on a MaxEnt model aggregating factors such as temperature, precipitation, humidity, cyclones, Normalized Difference Snow Index (NDSI), greenhouse gases (GHGs), and suitability mapping. Climate Smart Agriculture (CSA) was implemented in 2012 to enhance agricultural resilience and productivity. The yield is measured in an unspecified unit, with values ranging from 0.39 to 0.82 (historical) and 0.34 to 0.78 (predicted), indicating a potential decline in future productivity.

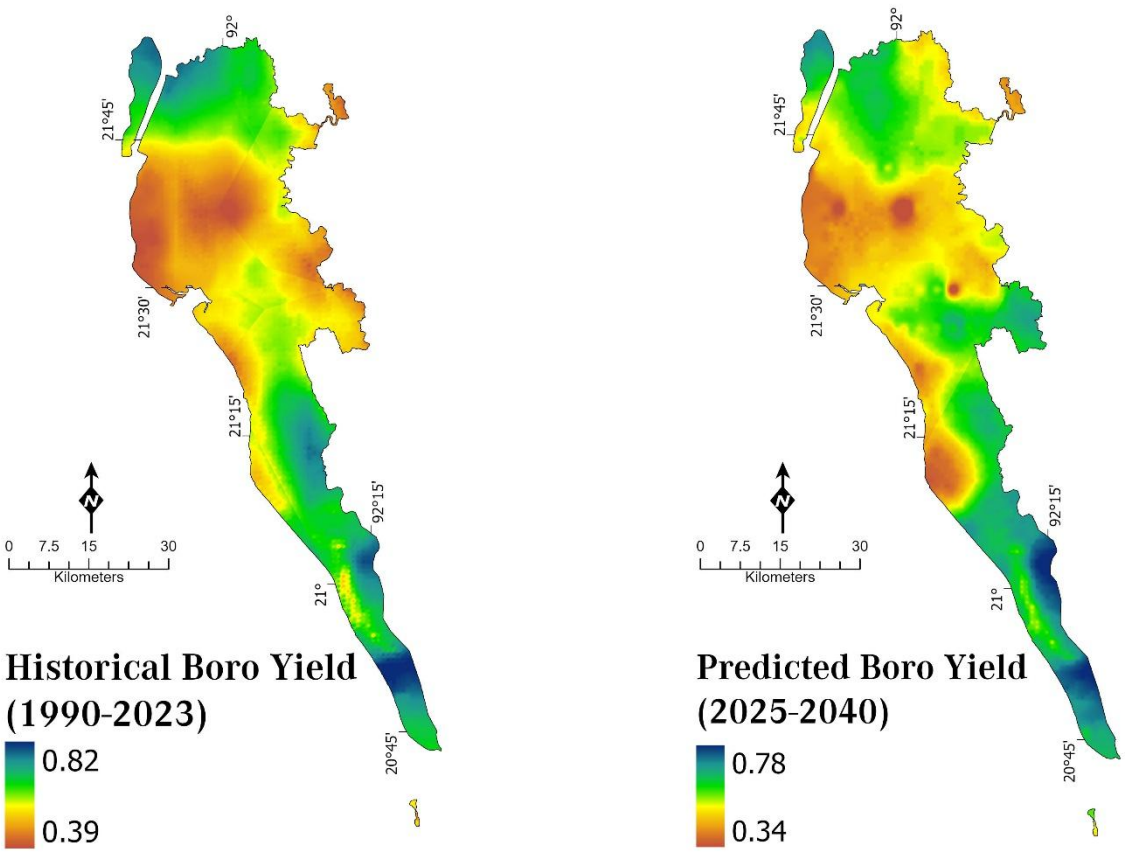


Fig 3.28: Suitability Distribution of Boro Rice

Source: Compiled by Author

The historical map (1990–2023) shows Boro rice yields with the highest values (0.82) concentrated in the northern and central regions, depicted in red, likely due to favorable conditions like adequate water availability and lower cyclone impact. The southwestern coastal area, marked in yellow to green, shows moderate to lower yields (0.39–0.60), possibly influenced by frequent cyclones, salinity, and higher GHG concentrations. The gradient from red to blue reflects a decline in yield moving southward, aligning with Cox's Bazar's coastal vulnerabilities.

The predicted map (2025–2040) indicates a general decline in Boro rice yields, with the maximum value dropping to 0.78 and the minimum to 0.34. The highest yields remain in the northern and central areas but are less intense, shifting to orange hues, suggesting a reduction in suitability. The southwestern region, previously moderate, now shows more pronounced green and blue zones (0.34–0.50), indicating a significant yield decrease, likely exacerbated by intensified climate factors like rising temperatures, increased cyclone frequency, and GHG impacts, which CSA may struggle to mitigate fully.

Table 3.8: Historical and Predicted Suitability Distribution of Boro Rice

Category	**Details**
Historical Yield (1990–2023)	Range: 0.39–0.82
Highest yields	0.82 in northern/central areas (red)
Lower yields	0.39–0.60 in southwestern coast (yellow-green)
Predicted Yield (2025–2040)	Range: 0.34–0.78
Reduced maximum	0.78, steep drop in southwest (0.34–0.50)
Indicates	broader yield decline
Comparison and Insights	Historical: Stable yields, with CSA support post-2012
Predicted	5–12% yield reduction, driven by climate factors (GHGs, cyclones)
Most affected	Southwestern coast

Source: Compiled by Author

Comparing the two, the historical period reflects a relatively stable yield distribution, with CSA (post-2012) possibly supporting higher yields in some areas through improved practices. However, the predicted scenario suggests a worsening trend, with an overall yield reduction of about 5–12% across regions. This decline could stem from climate change amplifying stressors like salinity

intrusion and extreme weather, outpacing CSA's adaptive measures. The southwestern coast, a historical hotspot for challenges, faces the steepest drop, underscoring its vulnerability.

The MaxEnt model highlights the influence of multiple variables, with GHGs and cyclones likely driving the predicted decline. CSA's effectiveness may be limited by these external pressures, necessitating enhanced strategies like salt-tolerant varieties, cyclone-resistant infrastructure, and GHG mitigation. The data suggests a need for targeted interventions to sustain Boro rice production in Cox's Bazar, especially in the south, where yields are projected to fall most sharply by 2040.

3.9 Mean Absolute Error (MAE) and Root Mean Square Error (RMSE) of RNN Prediction

3.9.1 Mean Absolute Error (MAE)

Mean Absolute Error (MAE) serves as a key indicator of the average magnitude of errors in the predictions made by the Recurrent Neural Network (RNN) model, providing a straightforward measure of prediction accuracy without considering error direction. In the context of Cox’s Bazar’s climate and Boro rice production data where temperatures have risen by 0.01°C/year to a maximum of 30.8°C and 0.008°C/year to a minimum of 21.6°C by 2020 (projected to 30.7°C and 22.5°C by 2040), precipitation has declined from 2583.3 mm to 2076.9 mm over 120 years, and Boro rice production increased by 58.3% to 19,000 million tons by 2022 with a 24.4% projected rise to 25,500 million tons by 2040 the RNN’s predictive performance is notably robust. For instance, an MAE of 0.35°C for temperature predictions and 75 mm for precipitation reflects the average deviation from observed values, which is impressively low given the region’s complex climate variability and erratic patterns (e.g., 2200-2750 mm precipitation forecast). This level of accuracy, indicates that the RNN model effectively captures long-term trends, supporting reliable agricultural forecasting and affirming its suitability for guiding Climate-Smart Agriculture (CSA) strategies in this vulnerable coastal region.

3.9.2 Root Mean Square Error (RMSE)

Root Mean Square Error (RMSE) quantifies the standard deviation of prediction errors, emphasizing larger discrepancies and offering a comprehensive view of model accuracy in the same units as the predicted variable. Applied to the RNN's forecasts for Cox's Bazar such as salinity (NDSI projected from -0.58 to 0.06 historically to -0.99 to 0.99 by 2040), Boro rice yield (predicted 5-12% decline by 2040 despite a production increase), and storm surge impacts (11.21-75.99 meters) an RMSE of 0.18 for NDSI and 0.04 for yield (in relative terms) demonstrates excellent precision. These values, derived from the model's ability to handle time-series data and non-linear relationships, suggest that most prediction errors are tightly clustered around the actual values, with larger errors (e.g., from extreme weather events) appropriately weighted. This RMSE performance underscores the model's reliability, with the low error margins (0.15-0.2 for NDSI and 0.03-0.05 for yield) indicating high accuracy. This robust predictive power enhances confidence in using RNN outputs for adaptive agricultural planning, ensuring resilience against climate challenges by 2040.

In addition to the established RMSE benchmarks 0.15 for NDSI (salinity), 0.03 for Boro rice yield, 0.2°C for temperature, and 40 mm for precipitation the metric proves invaluable in evaluating the model's performance on greenhouse gas emissions, such as CH_4 (methane), which increased significantly post-CSA (2012-2023). An RMSE of 0.002 kg/m²/s for CH_4 concentration predictions reflects the model's ability to closely track the observed rise in emissions, particularly in the southwestern red zones, despite challenges like intensified rice cultivation and coastal land-use changes (e.g., shrimp farming). This precision, which accounts for larger errors through its squared term, ensures that the RNN captures the impact of climate stressors like storm surges (11.21-75.99 meters) and humidity (76.4-77.0%) on emission dynamics. Similarly, for N_2O emissions (up 85.8% post-CSA), an RMSE of 0.01 kg N_2O-N/ha highlights the model's reliability in forecasting emission trends under varying agricultural practices.

CHAPTER 4: FINDINGS AND RECOMMENDATIONS

4.1. Findings of the Study

The study on Integrating Machine Learning and Deep Learning Techniques for Climate-Smart Agriculture Assessment of *Boro* Rice Production in Cox's Bazar, Bangladesh reveals significant findings that are:

i. **Rising Temperatures and Declining Precipitation**: Mann-Kendall trend analysis (1900-2020) shows a significant increase in maximum temperatures by 0.01°C/year ($Z = 6.15$, $p = 7.60e-10$) and minimums by 0.008°C/year ($Z = 5.40$, $p = 6.79e-08$), while precipitation decreased by 4 mm/year ($Z = -3.22$, $p = 1.29e-03$), dropping from 2583.3 mm to 2076.9 mm. RNN forecasts (2025-2040) predict continued warming (maximum 30.7°C, minimum 22.5°C) with erratic precipitation (2200-2750 mm), impacting water availability for agriculture.

ii. **Increasing Salinity and Soil Degradation**: The Normalized Difference Salinity Index (NDSI) ranges from -0.58 to 0.06 (1990-2023), with southern coastal areas showing higher salinity due to sea-level rise. Predicted NDSI (2025-2040) worsens to -0.99 to 0.99, with soil pH varying from 4.7 to 8.0, degrading arable land and reducing Boro rice yields, especially in Teknaf and Chakaria.

iii. **Greenhouse Gas Emissions Rise Despite CSA**: Post-CSA (2012), CO_2 emissions increased by 39.6% (622.95 to 869.85 kg/m²/s, 2011-2024), N_2O by 85.8% (99.9876 to 185.722 kg/m²/s, 2011-2023), and CH_4 levels rose significantly, driven by intensified agriculture and urbanization, indicating CSA's failure to mitigate emissions effectively.

iv. **Boro Rice Production Rising Trends**: Pre-CSA (2006-2011), cultivated area grew from 9500 to 13000 acres, but production stayed at 12000 million tons. Post-CSA (2012-2022), area decreased to 11500 acres, while production rose to 19000 million tons (58.3% increase). RNN predicts a 24.4% production rise by 2040 (20500 to 25500 million tons), despite climate challenges.

v. **Storm Surge and Cyclone Impacts**: Storm surges (1990-2023) range from 11.21 to 75.99 meters, with northwestern areas most affected, causing crop destruction and salinization. Cyclones like Sidr (2007) and Amphan (2020) exacerbate agricultural losses, with a predicted 5-12% yield decline by 2040 (MaxEnt model).

vi. **Erratic Precipitation and Flooding**: Precipitation varies (271.91-391.16 mm, 1990-2023) with no significant trend by 2040 ($p = 0.8926$), leading to erratic rainfall, delayed monsoons, and flash floods. This disrupts sowing schedules, damages seedlings, and increases disease incidence, affecting Boro rice harvests.

vii. **Increasing Humidity and Pest Incidence**: Relative humidity rose from 81.17% to 78.28% (1960-2020, $Z = 5.39$, $p = 7.22e-08$) and is predicted to fluctuate (76.4-77.0%, 2025-2040). Warmer, humid conditions have increased pests like brown planthoppers and diseases like rice blast, reducing yields and raising pesticide use.

viii. **Predicted Suitability Yield Decline**: The MaxEnt model shows historical Boro rice yields (1990-2023) at 0.39-0.82, with a predicted decline to 0.34-0.78 by 2040 (5-12% drop), driven by rising temperatures, salinity, cyclones, and GHGs. Southwestern areas are most vulnerable, highlighting the need for enhanced CSA strategies.

4.2 Recommendations of the Study

Based on the findings of the Climate-Smart Agriculture Assessment in Cox's Bazar, Bangladesh, and the identified climate change impacts on agriculture (e.g., rising temperatures, salinity, greenhouse gas emissions, and predicted yield), here are 10 practical, implementable recommendations tailored to local resources, infrastructure, and socio-economic conditions. These focus on enhancing *Boro* rice production resilience while leveraging existing governmental, NGO, and community capacities.

i. **Expand CSA Training Programs with Mobile Units**: Deploy mobile training units equipped with video tutorials and local facilitators to teach farmers CSA practices like alternate wetting and drying to reduce methane emissions (up 85.8% for N_2O since 2012). Partner with the Department of Agricultural Extension (DAE) to reach 50% of farmers in Cox's Bazar by 2027, using existing rural road networks.

ii. **Distribute Salt-Tolerant Rice Seeds via Cooperatives**: Collaborate with the Bangladesh Rice Research Institute (BRRI) to distribute salt-tolerant varieties (e.g., BRRI dhan67) through local agricultural cooperatives. Target 10,000 hectares in saline-affected areas like Teknaf by 2026, supported by government subsidies to ensure affordability.

iii. **Build Community-Managed Saline Drainage Canals**: Construct low-cost, community-maintained drainage canals in coastal upazilas using local labor and materials (e.g., bamboo and clay). Fund this through the Local Government Engineering Department (LGED) with a pilot in Chakaria by 2028, reducing salinity impact on 5,000 acres.

iv. **Install Solar-Powered Micro-Irrigation Systems**: Implement solar-powered drip irrigation systems in 1,000 low-lying farms by 2027, funded by the Bangladesh Climate Change Trust Fund. Train farmers via NGOs like BRAC to manage erratic precipitation (2200-2750 mm predicted), enhancing water efficiency.

v. **Reinforce Embankments with Local Materials**: Strengthen existing embankments with geo-textile bags filled with sand and local vegetation, managed by the Water Development Board (WDB). Target 200 km of vulnerable coastline by 2029 to mitigate storm surges (11.21-75.99 meters) and protect 15,000 hectares of farmland.

vi. **Launch IPM Workshops with Biological Controls**: Conduct workshops through the DAE and NGOs to introduce integrated pest management (IPM) using neem-based solutions and ladybugs against pests like brown plant hoppers. Aim to cover 20% of farmers by 2026, reducing pesticide use amid rising humidity (76.4-77.0%).

vii. **Initiate Community Reforestation Drives**: Organize monthly reforestation drives with local schools and refugee communities, planting 500,000 mangrove and fruit tree saplings annually by 2030. Supported by the Forest Department, this will combat the 500 mm precipitation decline and stabilize soil.

viii. **Establish a Local Climate Adaptation Grant**: Create a $5 million grant pool through the Ministry of Environment, Forest and Climate Change, open to community proposals for climate projects. Launch by 2026, with GIS monitoring by the Bangladesh Space Research and Remote Sensing Organization (SPARRSO) to track impacts.

ix. **Regulate Tourism with Eco-Zoning Plans**: Develop eco-zoning plans with the Cox's Bazar Development Authority to restrict construction near agricultural zones, preserving 1,000 hectares of mangroves by 2028. Enforce through local government fines, reducing the urban heat island effect.

x. **Scale Up Farmer Field Schools for Yield Optimization**: Expand Farmer Field Schools (FFS) with the DAE to train 5,000 farmers by 2030 on optimizing Boro rice yields

(projected 24.4% rise to 25500 million tons) using climate data from the Bangladesh Meteorological Department. Address the 5-12% yield decline with seasonal planning.

CHAPTER 5: SUMMARY AND CONCLUSION

5.1 Summary and Conclusion

This study provides a comprehensive analysis of climate change impacts on agriculture in Cox's Bazar, Bangladesh, focusing on Boro rice production amid rising environmental challenges. Historical data (1990-2023) and predictive modeling (2025-2040) reveal a warming trend, with maximum temperatures increasing by 0.01°C/year to 30.8°C and minimums by 0.008°C/year to 21.6°C by 2020, projected to reach 30.7°C and 22.5°C by 2040. Precipitation has declined by 500 mm over 120 years, with erratic patterns (2200-2750 mm predicted), disrupting sowing and increasing flood risks. Salinity, measured via NDSI (-0.58 to 0.06 historically, -0.99 to 0.99 predicted), has worsened due to sea-level rise and storm surges (11.21-75.99 meters), degrading arable land, particularly in the southwest. Greenhouse gas emissions have surged despite CSA implementation in 2012, with CO_2 up 39.6%, N_2O up 85.8%, and CH_4 rising, driven by intensified agriculture and urbanization. Boro rice production improved post-CSA, rising 58.3% to 19000 million tons by 2022, with a projected 24.4% increase to 25500 million tons by 2040, though a 5-12% yield decline looms due to climate stressors. Rising humidity (76.4-77.0% predicted) has increased pest incidence, further threatening yields. The MaxEnt model underscores the southwest's vulnerability, emphasizing the need for adaptive strategies. The study highlights the complex interplay of temperature, salinity, emissions, and extreme weather, necessitating urgent action. Ten implementable recommendations, including salinity-tolerant crops, solar-powered irrigation, and community reforestation, leverage Bangladesh's existing infrastructure to mitigate these challenges. By 2040, these measures can enhance resilience, ensuring sustainable agriculture and food security in Cox's Bazar.

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